

# The haptic function

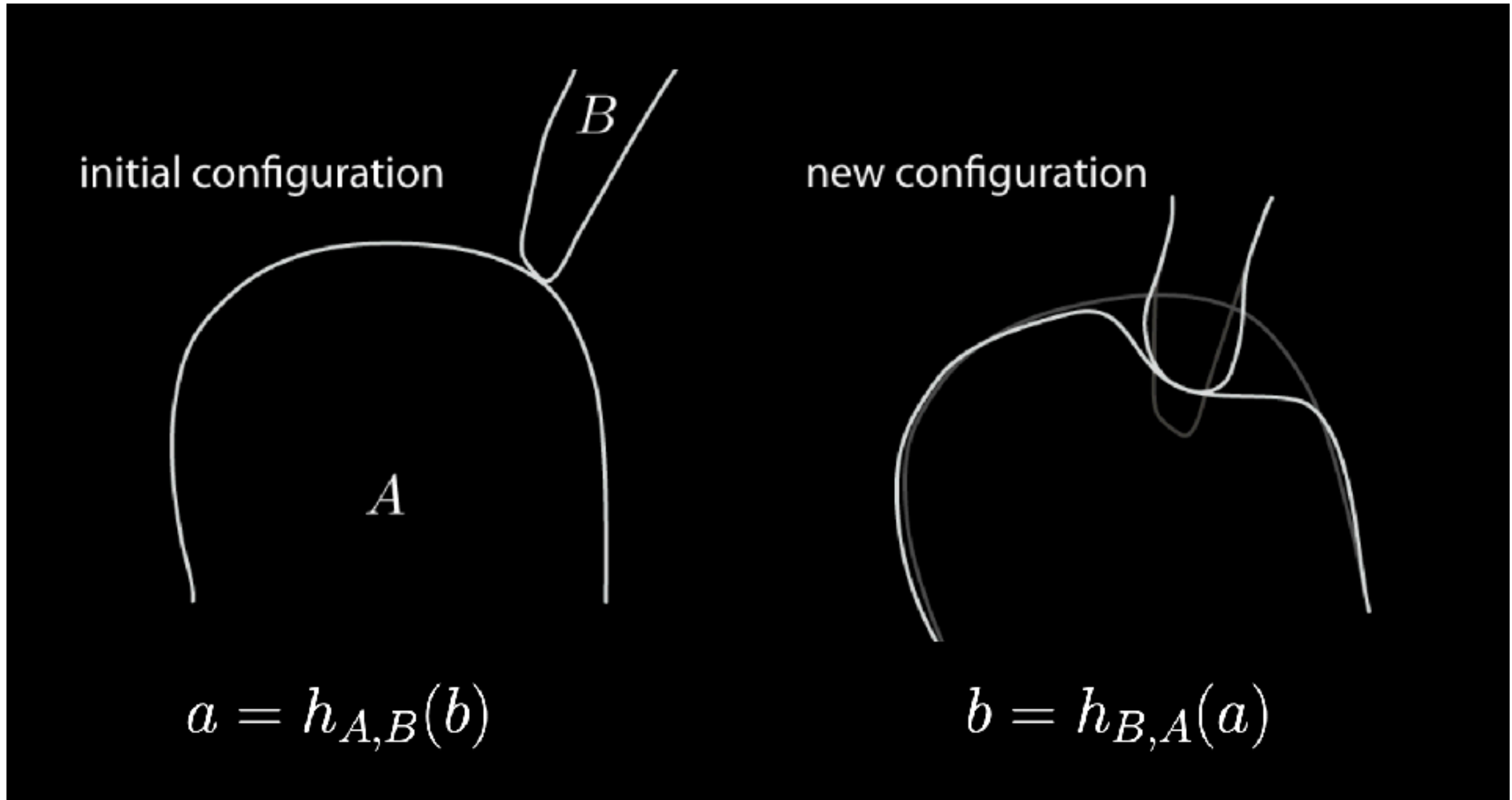
David Gueorguiev

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Sorbonne Université - 2025 - 2026

# Penhaptic function?

- Plenoptic function of the light field: 5 dimensions
- Penhaptic function: infinite dimensions.



# Motions for haptic exploration

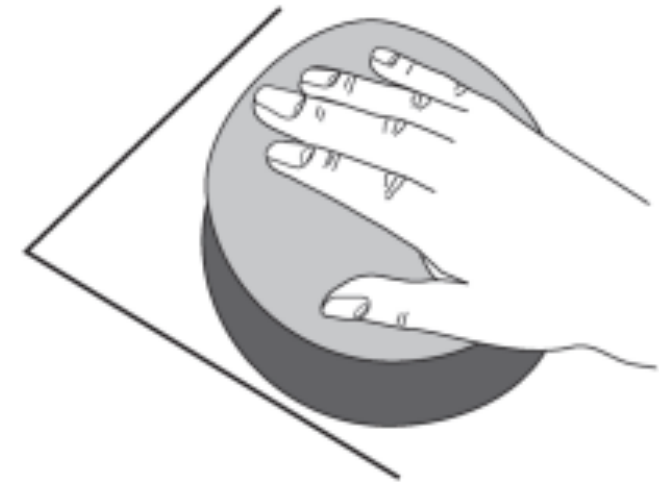
Lateral Motion:  
**Texture**



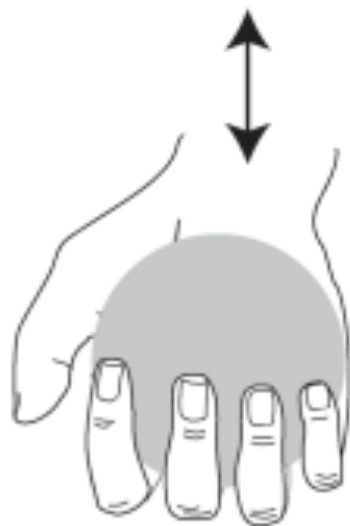
Pressure:  
**Hardness**



Static Contact:  
**Temperature**



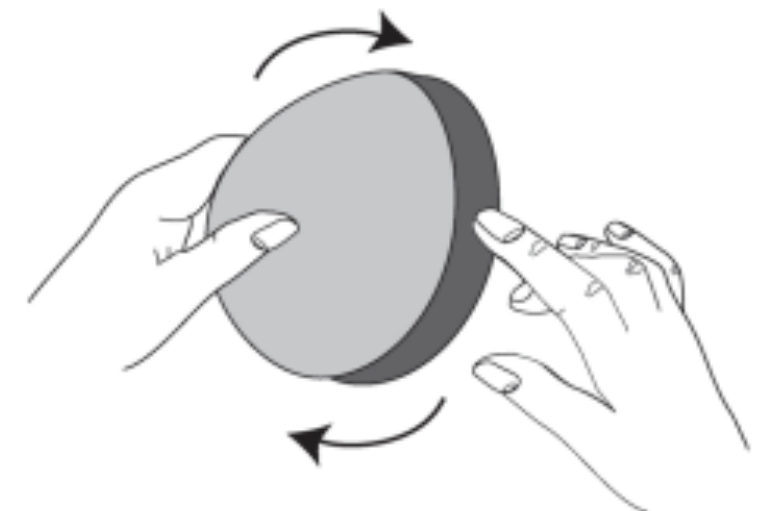
Unsupported  
Holding:  
**Weight**

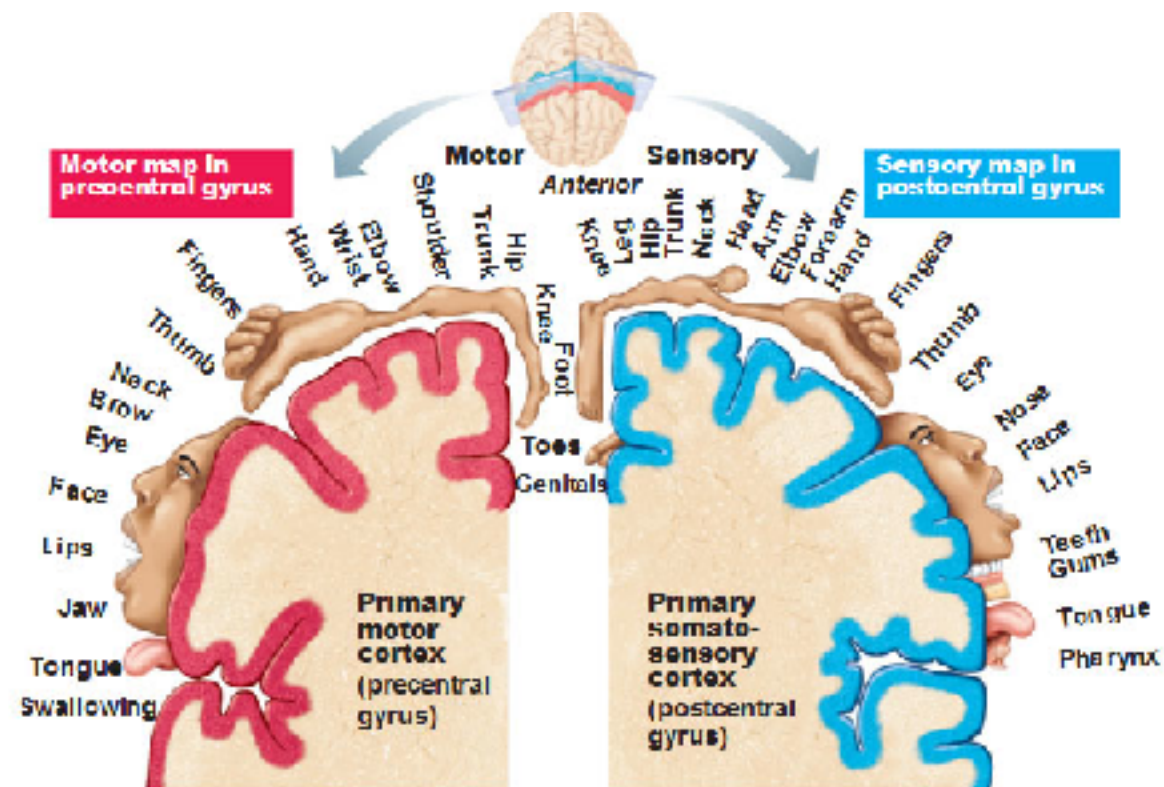


Enclosure:  
**Global Shape,  
Volume**

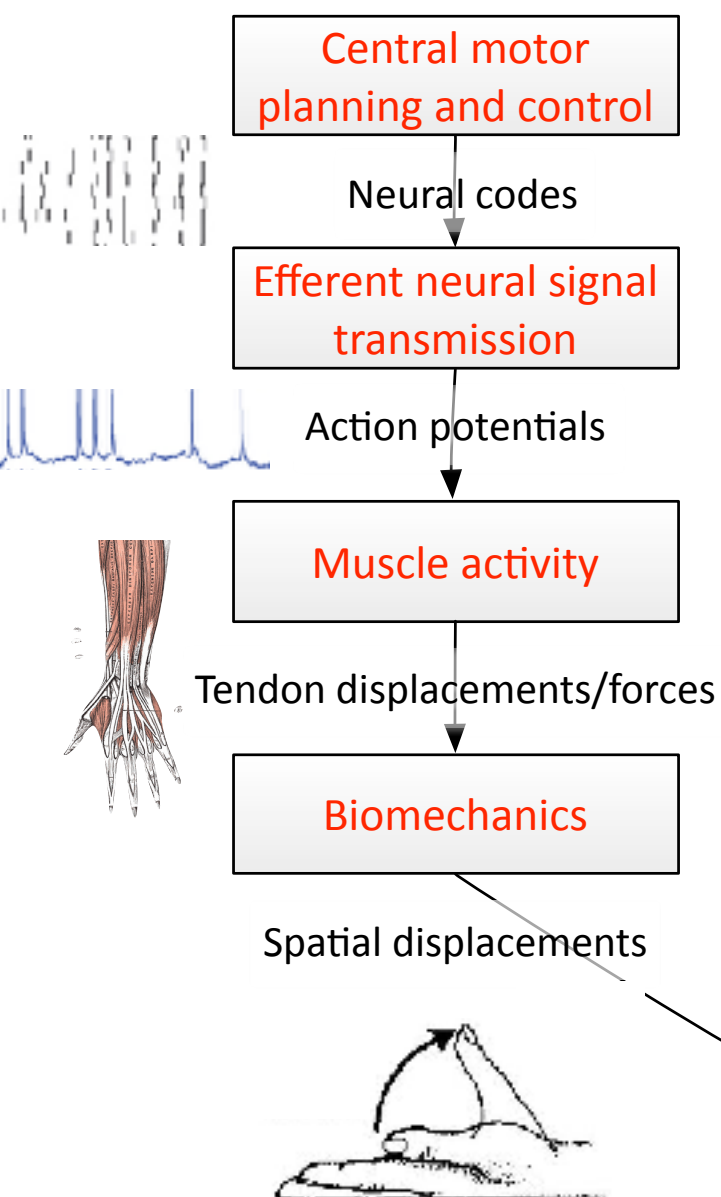


Contour Following:  
**Global Shape,  
Exact Shape**

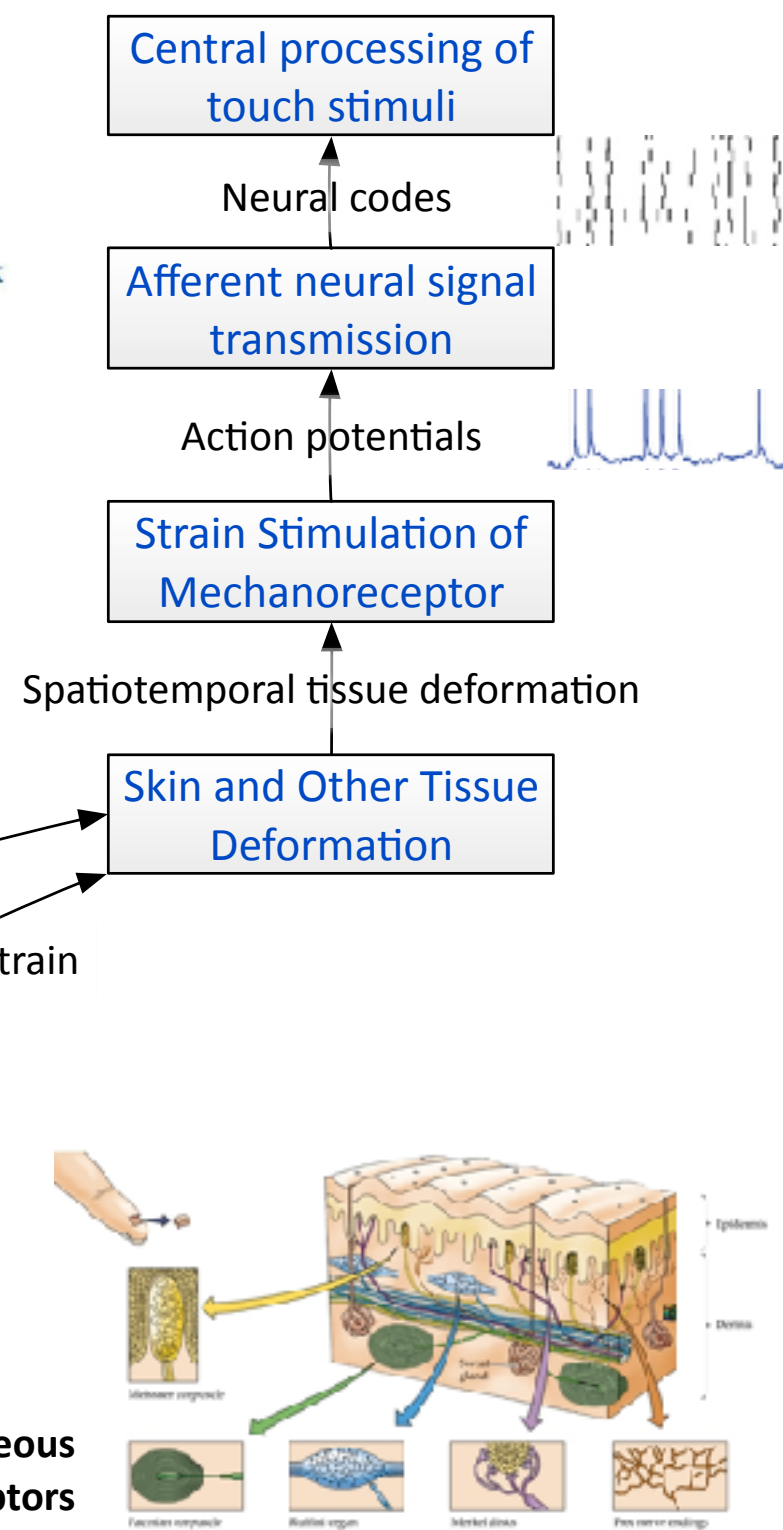




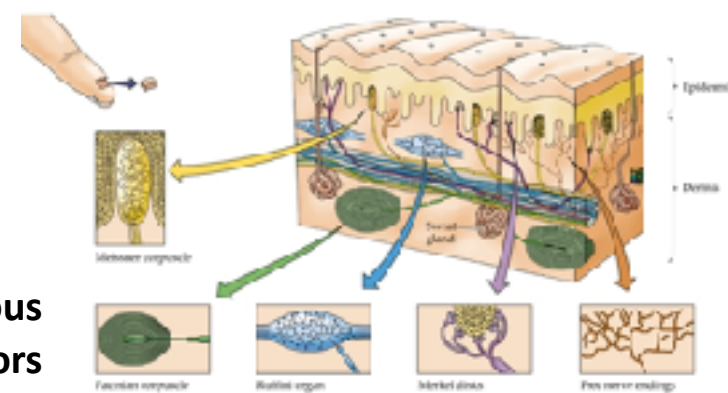
## Descending Pathway



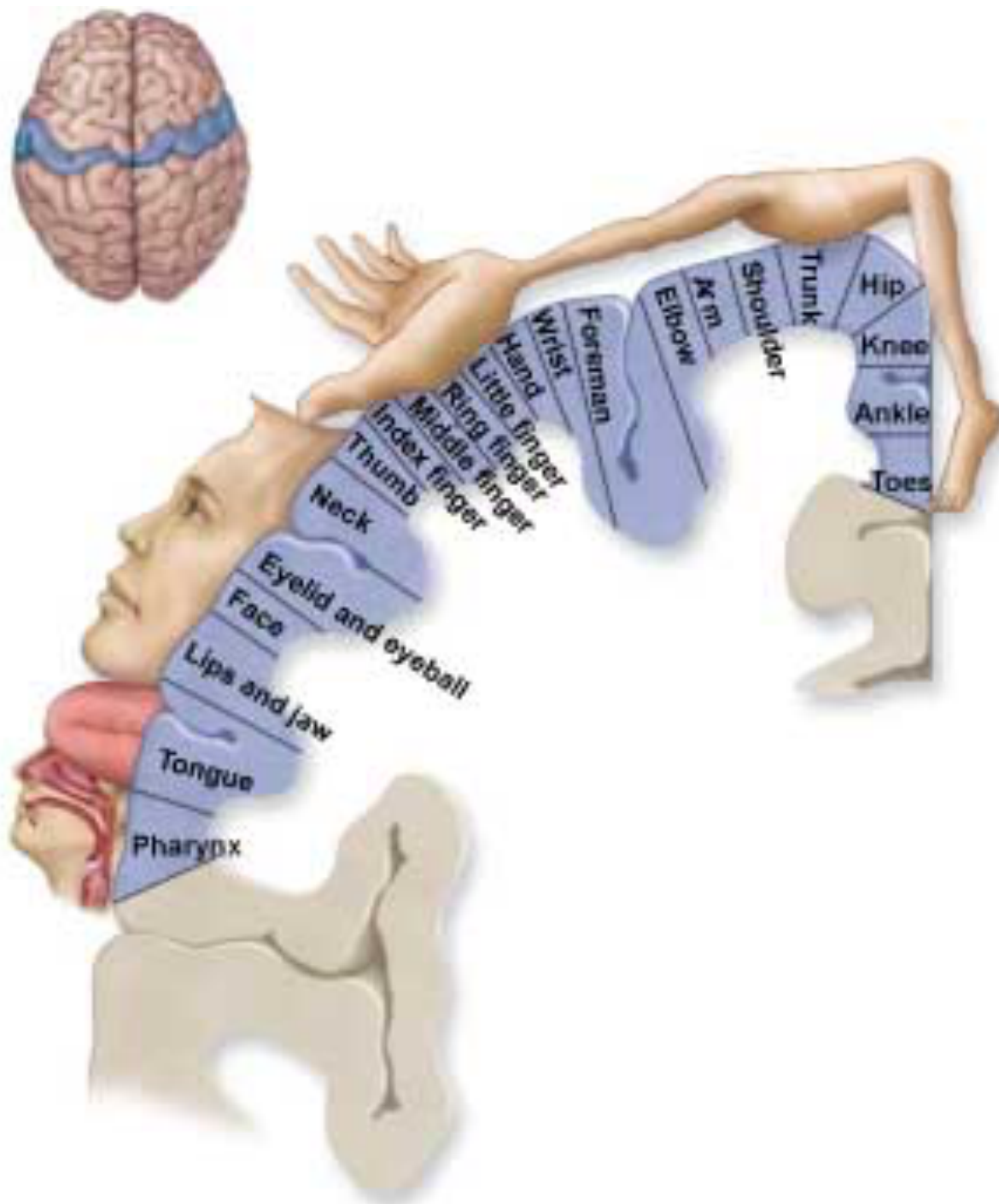
## Ascending Pathway



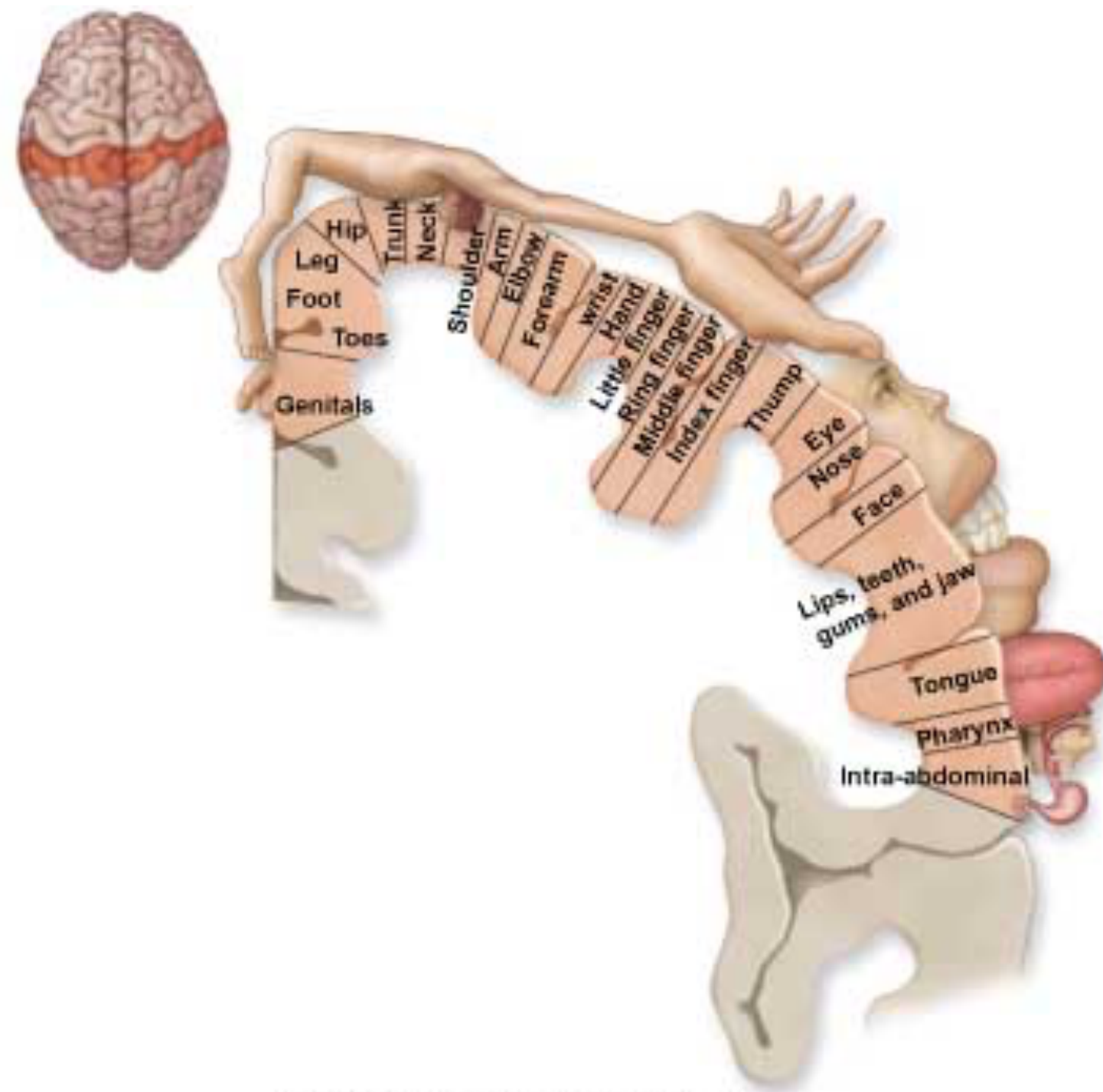
## Cutaneous Mechanoreceptors



# Homunculus somatotopic organisation



a. Primary motor area



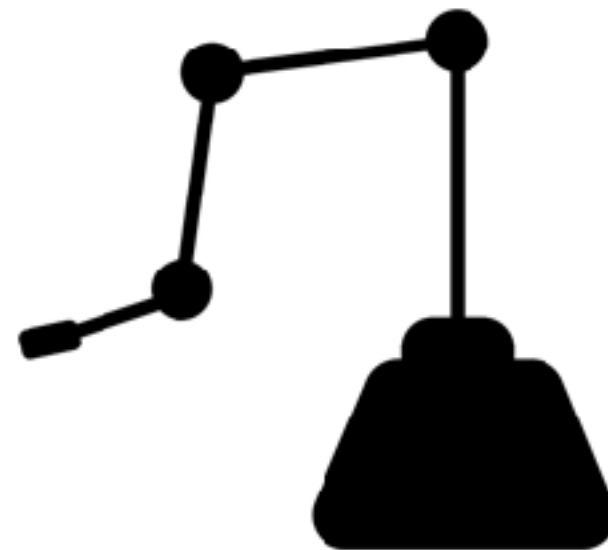
b . Primary somatosensory area

# **Perception and psychophysics**

**Which change in intensity produces a sensation?**

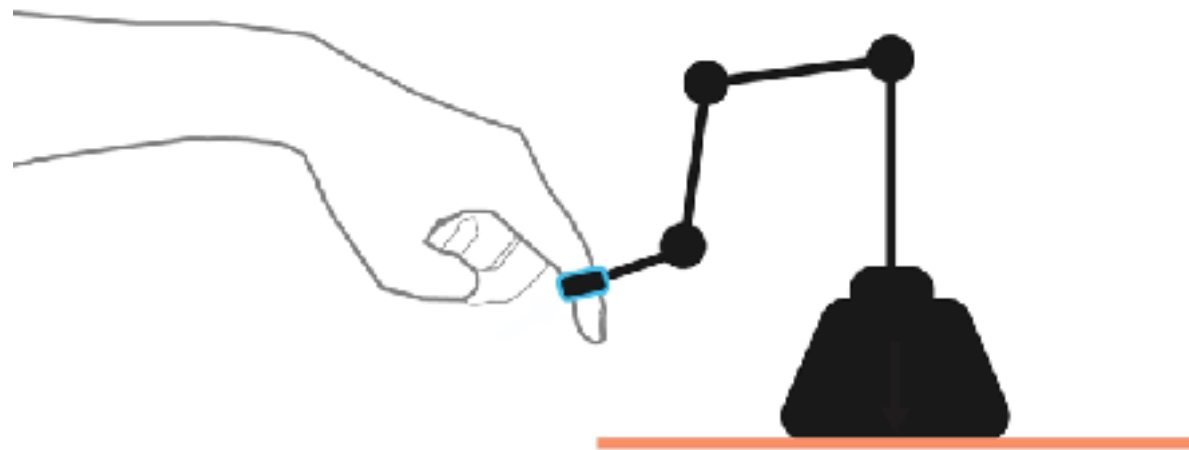
**How to quantify it?**

Research on human touch is a major gateway for the progress of haptics and the conception of novel haptic devices.



In turn, haptic devices can help study human touch as they can:

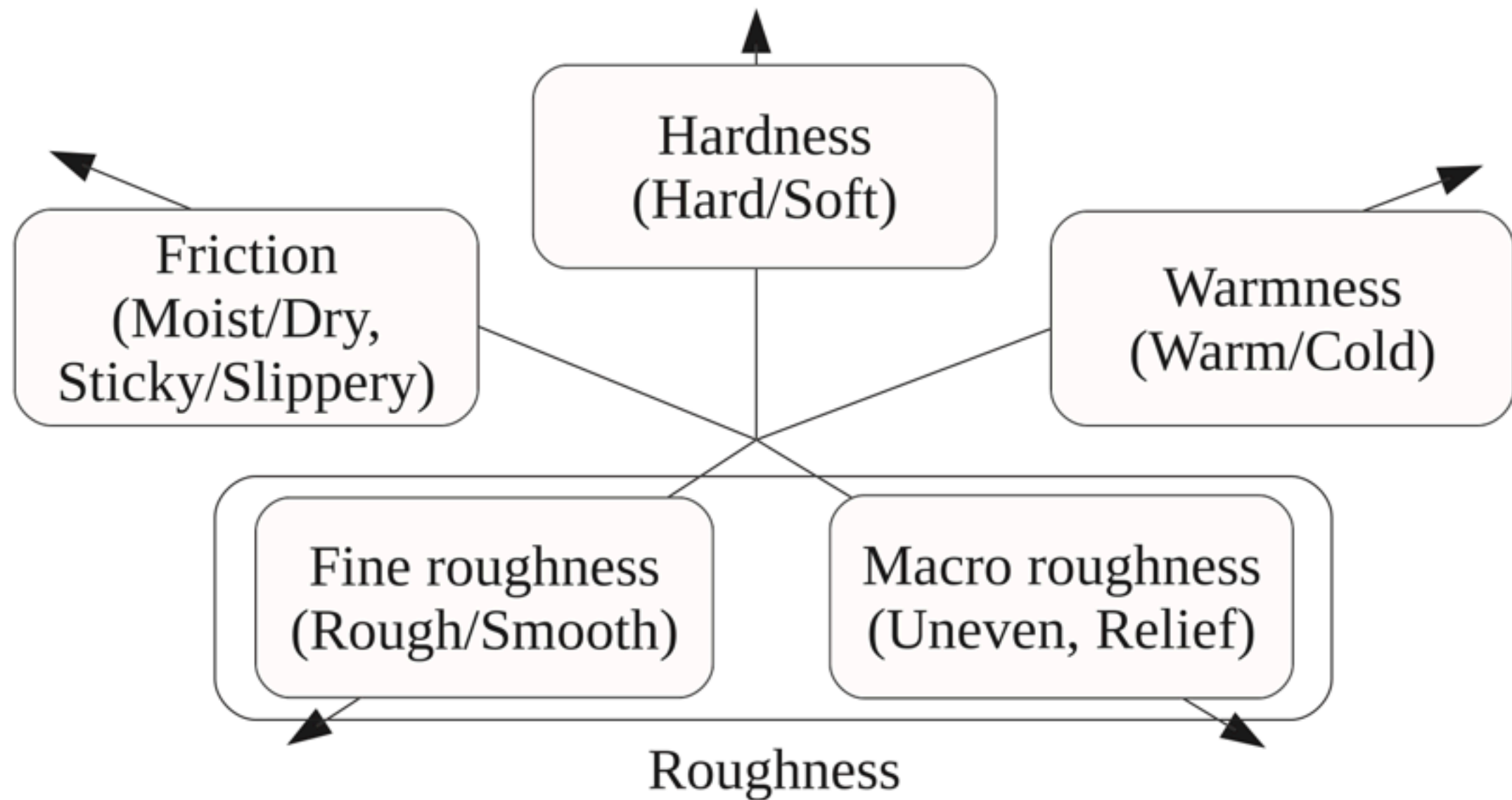
- precisely control stimuli and measure user motion and/or force.
- create stimuli that do not exist in the physical world.



# Perception studies

To study human touch perception, researchers rely on a set of established procedures that are known as psychophysical methods.

# Psychophysical tactile dimensions



# The origins of Psychophysics



1830: Ernst Weber: *First experiments on touch and light*



1860: Gustav Fechner: *Elements of psychophysics*

Quantitative investigation of the mechanisms that detect and process information in the nervous system

# Weber law

$$dp = k \frac{dS}{S}$$

$dp$  = différence de sensation

$dS$  = différence de stimulus

$S$  = valeur de référence du stimulus

$k$  = constante de Weber

## Example

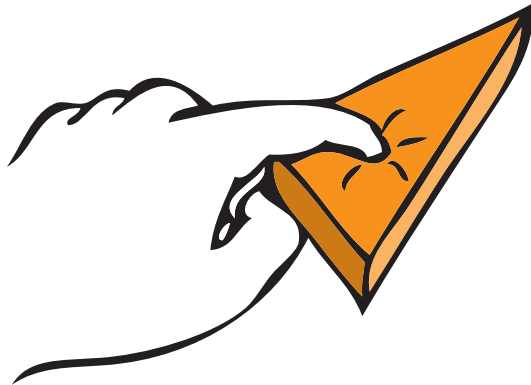
An item weights 100 grams, item B weights 105 grams

An item C weights 200 grams, item D weights 210 grams

Same perceived difference of sensation,  $dp$

# A few Weber fractions

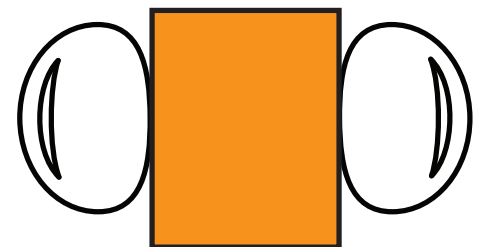
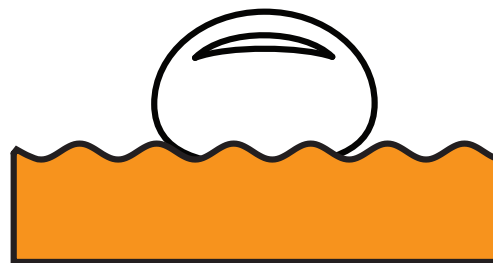
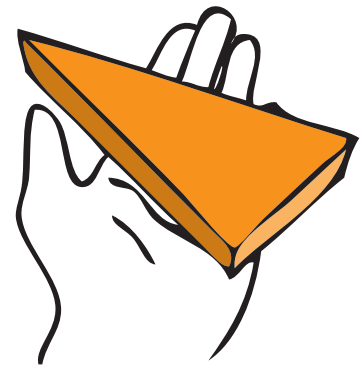
**Elasticity (23%)**



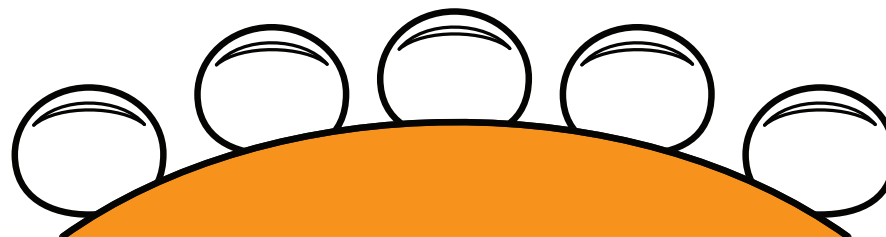
**Roughness:**  
amplitude (15%)  
frequency (9%)



**weight (7/9%)**



**Curvature  
(10%)**



# Human haptic sensing and output

**Table 3.2** Sensory thresholds of human hand. Unless otherwise stated, the values are for the index fingertip

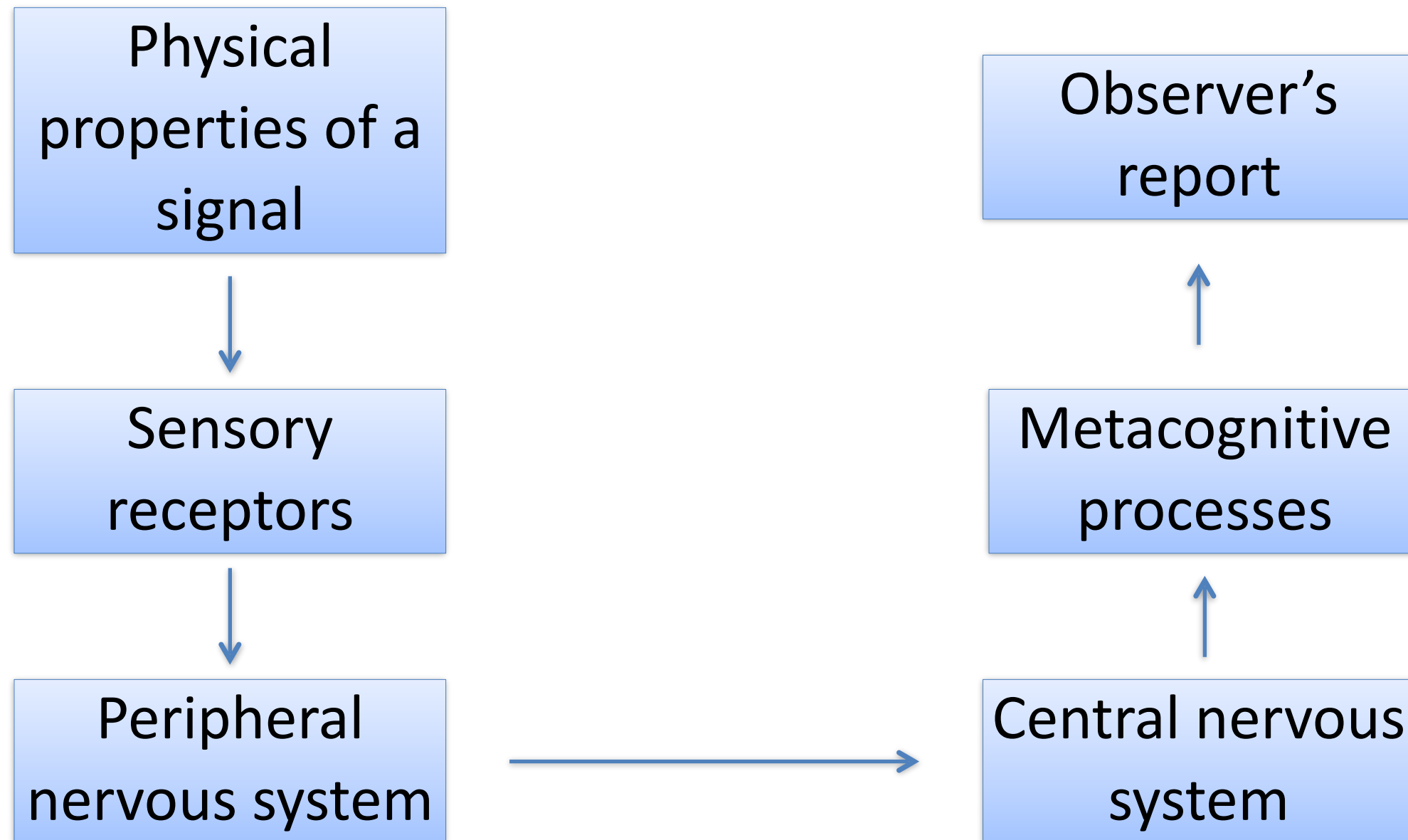
| Parameter           |                          | Value       | Notes                                    |
|---------------------|--------------------------|-------------|------------------------------------------|
| Force               | Absolute                 | 0.06 N      |                                          |
|                     | Differential             | 7 %         | Over a range of 2.5–10 N                 |
|                     |                          | 7 %         | Elbow muscle. Over a range of 25–400 N   |
|                     |                          | 15–27 %     | For forces less than 0.5 N               |
| Pressure            | Absolute                 | 0.019 g     | Women                                    |
|                     |                          | 0.055 g     | Men                                      |
| Friction            | Accurately scaled        | 0.43–2.79   | Range of friction coefficient            |
| Shear               | Accurately scaled        | 0.15–0.70 N |                                          |
| Vibration           | Intensity (differential) | 25 %        | Over a range of 10–20 dB sensation level |
|                     |                          | const 28 dB | For 0.4–3 Hz with 1 $\mu$ m peak         |
|                     | Temporal (absolute)      | –5 dB/dec   | For 3–30 Hz                              |
|                     |                          | –10 dB/dec  | For 30–250 Hz                            |
|                     |                          | +10 dB/dec  | For 250–700 Hz                           |
| Spatial resolution  |                          | 1.2 mm      | Discrimination of grating orientation    |
|                     |                          | 0.6 mm      | Detection of grating                     |
| Temporal resolution |                          | 5 ms        | Successiveness of mechanical pulses      |

**Table 3.4** Output capabilities of human

| Parameter |               | Value      | Notes                                                                       | Reference |
|-----------|---------------|------------|-----------------------------------------------------------------------------|-----------|
| Force     | Maximum       | 50/60 N    | Finger/wrist                                                                | [64]      |
|           |               | 100 N      | Elbow & shoulder                                                            |           |
|           | Control       | 11–15 %    |                                                                             | [61]      |
|           |               | 1 %        | Over 10 to 20 N                                                             | [64]      |
|           |               | 1 %        | Wrist, elbow & shoulder                                                     |           |
|           |               |            | Over 20 to 50 N                                                             |           |
|           |               | 10 %       | Over 5 to 18 N                                                              | [47]      |
|           | Bandwidth     | 2–3 Hz     |                                                                             | [55]      |
|           |               | 5 Hz       |                                                                             | [3]       |
| 2–6 Hz    |               |            | [26, 27]                                                                    |           |
| Motion    | Maximum speed | 17.6 rad/s |                                                                             | [17]      |
|           | Bandwidth     | <2 Hz      | Active touch for sensing                                                    | [36]      |
|           |               | 2–4 Hz     | Voluntary movements                                                         | [44, 51]  |
|           |               | 2–7 Hz     | Periodic tracking                                                           | [5]       |
|           |               | 4–8 Hz     | Skilled actions: hand writing, typing, tapping, playing musical instruments | [26, 36]  |
|           |               | 10 Hz      | Reflexive actions                                                           | [3, 4]    |
|           |               | <10 Hz     |                                                                             | [72]      |
|           |               | 8–12 Hz    | Finger tremor                                                               | [57]      |

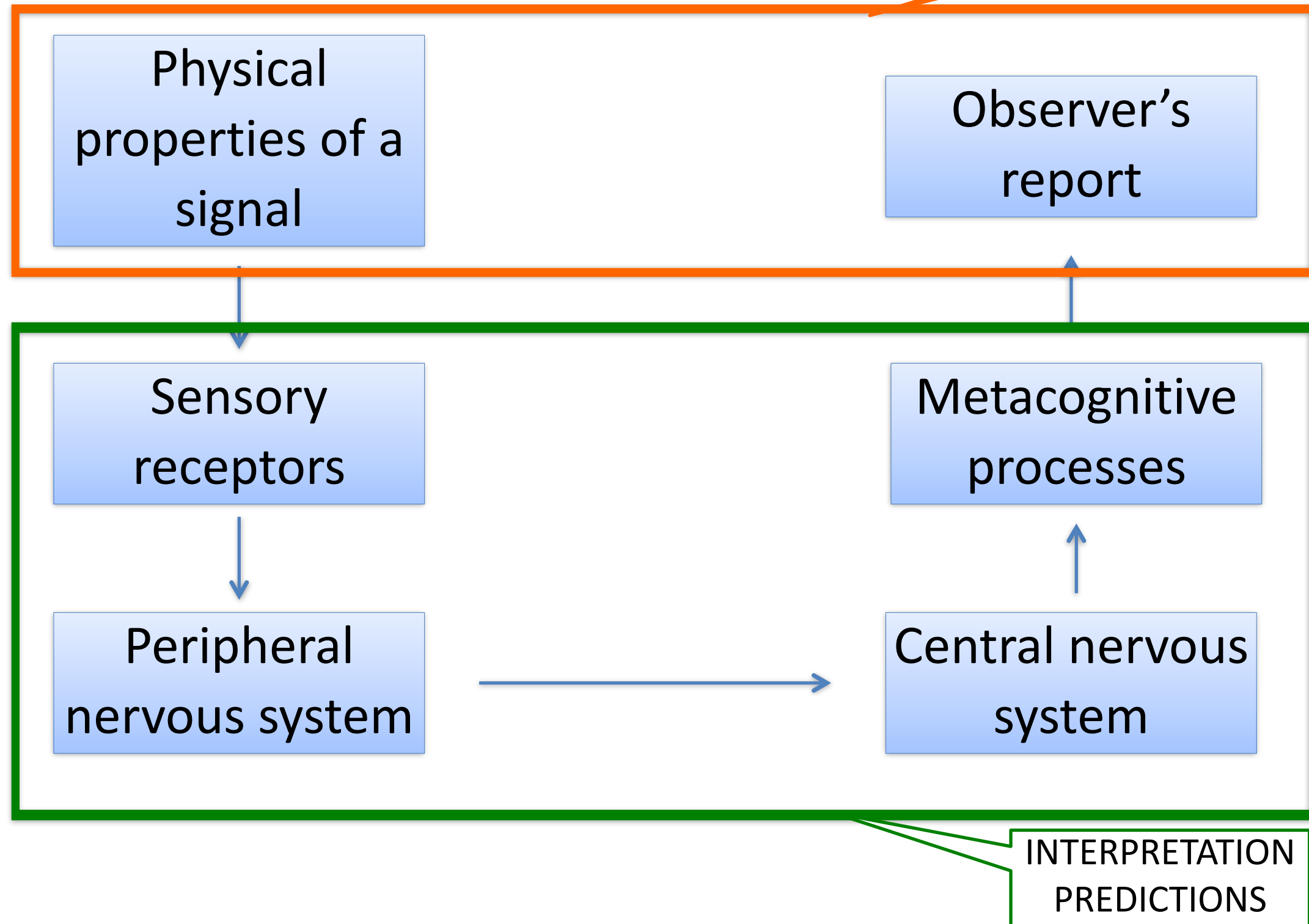
Samur, 2012

# Perceptual pathways



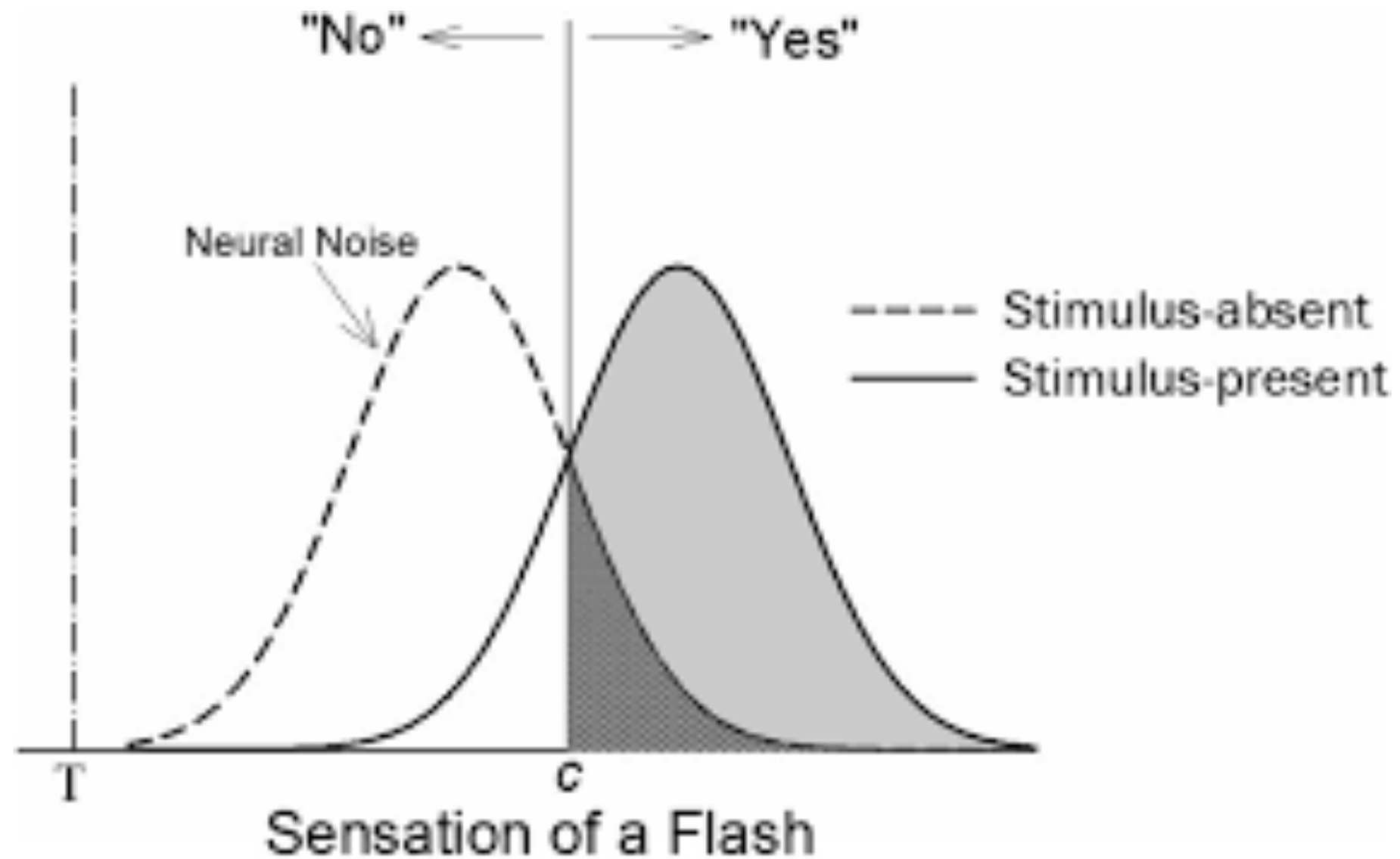
# Perceptual pathways

PSYCHOPHYSICAL  
EXPERIMENTS



INTERPRETATION  
PREDICTIONS

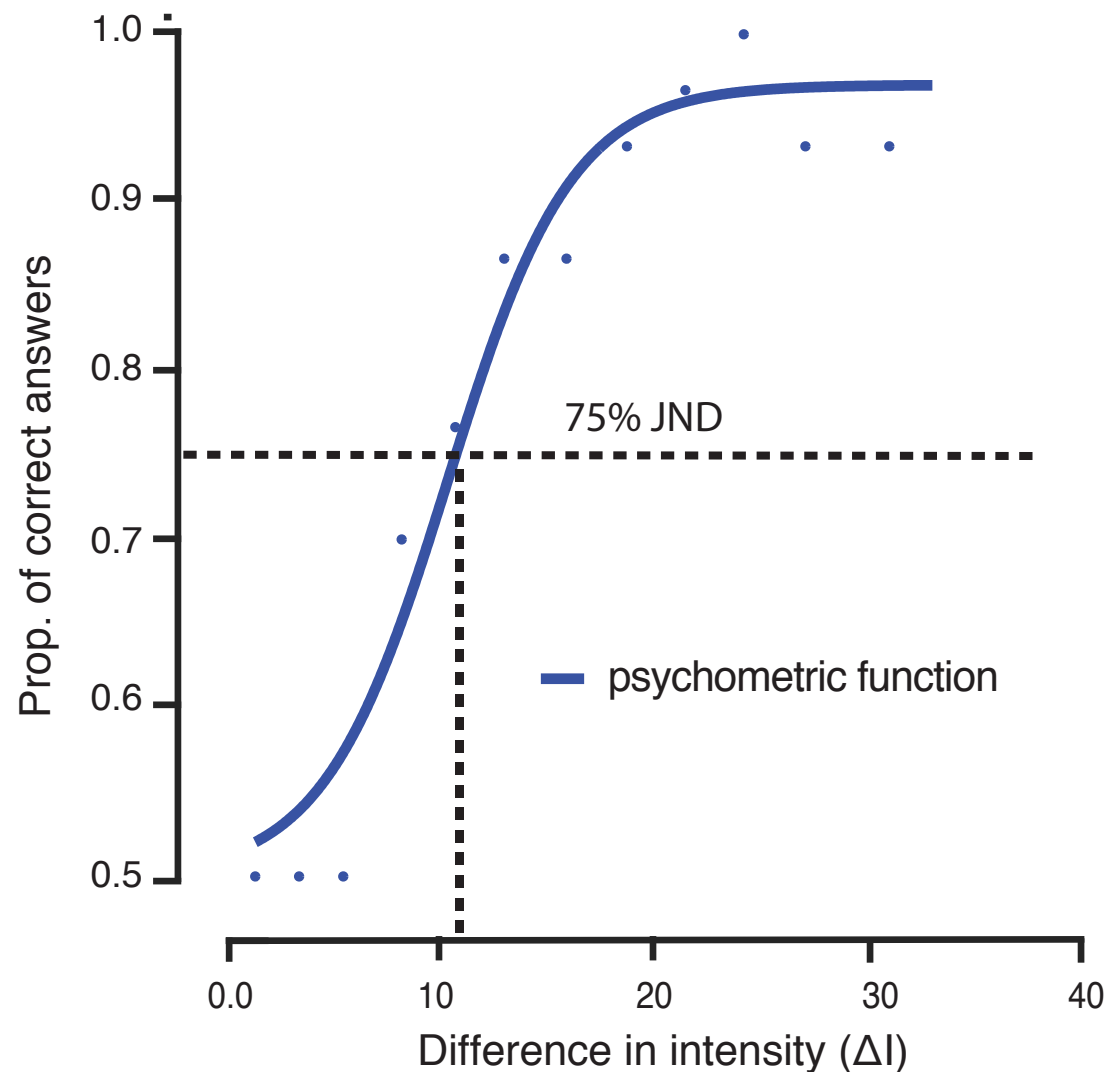
# Psychophysics are rooted in decision theory



# Psychometric function

The probability for humans to detect differences within a given feature of a physical stimulus, e.g. force, stiffness, weight etc.

**Here, the difference that is perceived 75% of the time is around 75%**



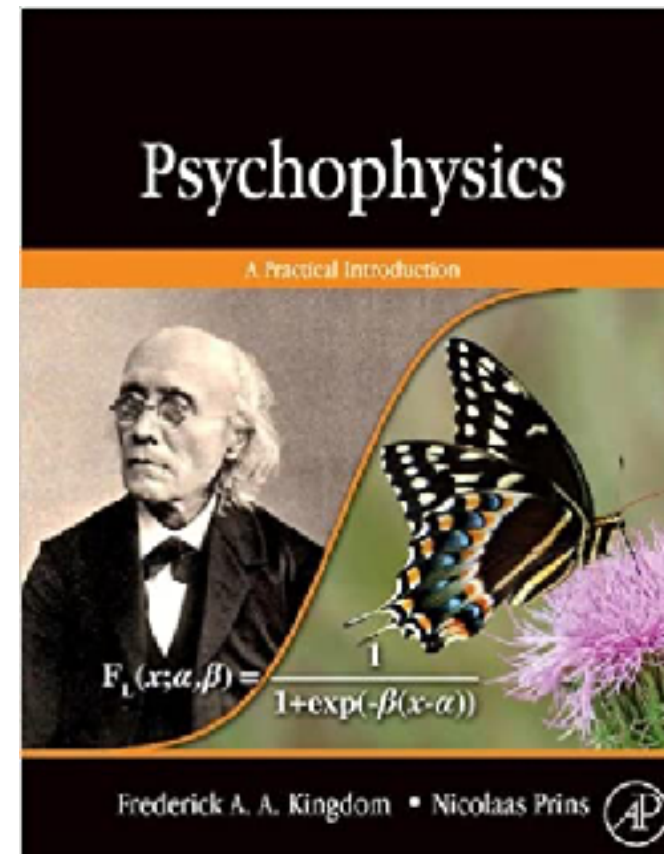
Just noticeable difference = sensory threshold (ex: 75% JND)

-> the difference between the changed stimulus and the reference is perceived 75% of the time.

# References for choosing and experimental method



Jones and Tan, 2013



Kingdom and Prins,  
several editions

# Three classical psychophysical procedures

Three classical methods (among many others)

- Method of the limits - fast but prone to bias
- Constant stimuli - robust but time-consuming
- Adaptive method - Optimized for JND tracking

# Method of the limits

- Increasing series
  - The stimulus is **increased** until detection
- decreasing series
  - The stimulus is **decreased** until unnoticed
- At least 4 series

*The sensory threshold is estimated as the average of the reversal points*

# The method of limits in practice

x = not perceived

# Ascending series

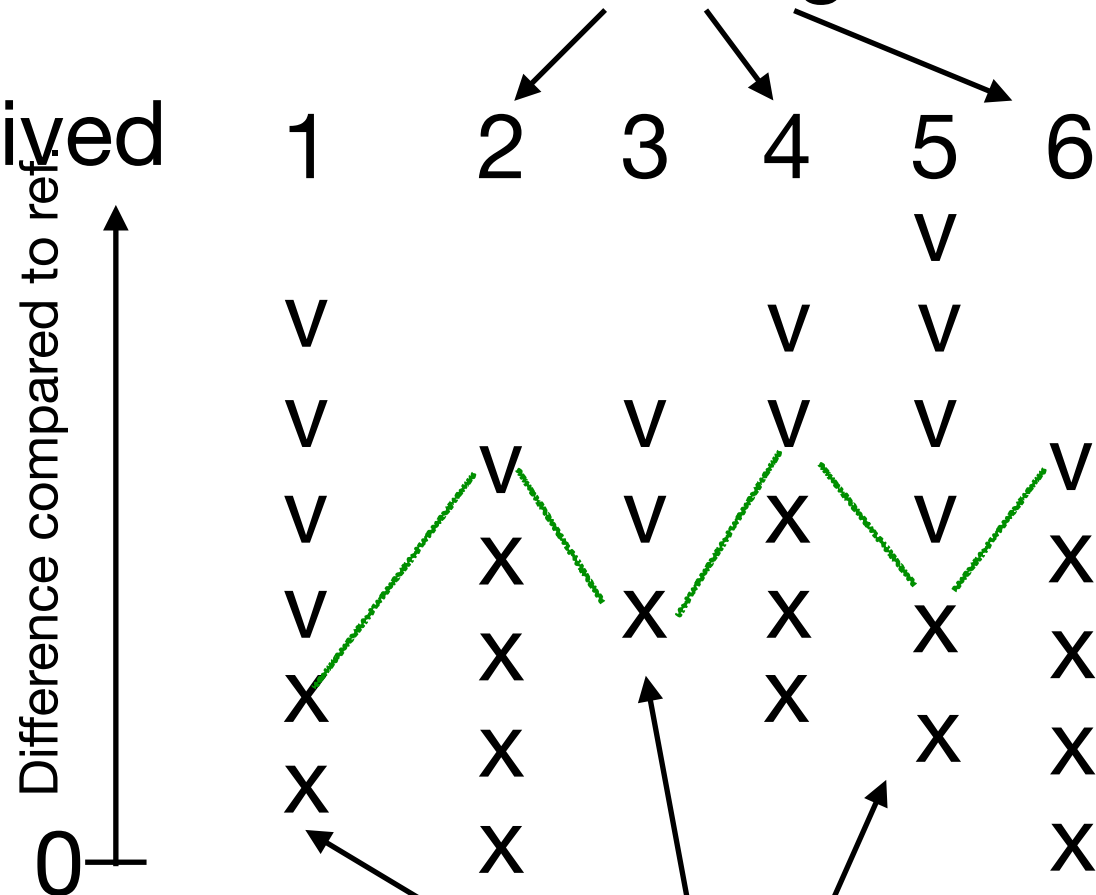
v = perceived

1 2 3 4 5 6

Ascending and descending series alternate to avoid habituation.

# Reversal points

# Descending series



# Method of the limits:

In practice, the method of the limits is often used to obtain a quick first estimate of the sensory threshold. This first estimate is then used to calibrate a more thorough method, usually a constant stimuli or adaptive.

## Fingerpad Skin Stretch Increases the Perception of Virtual Friction

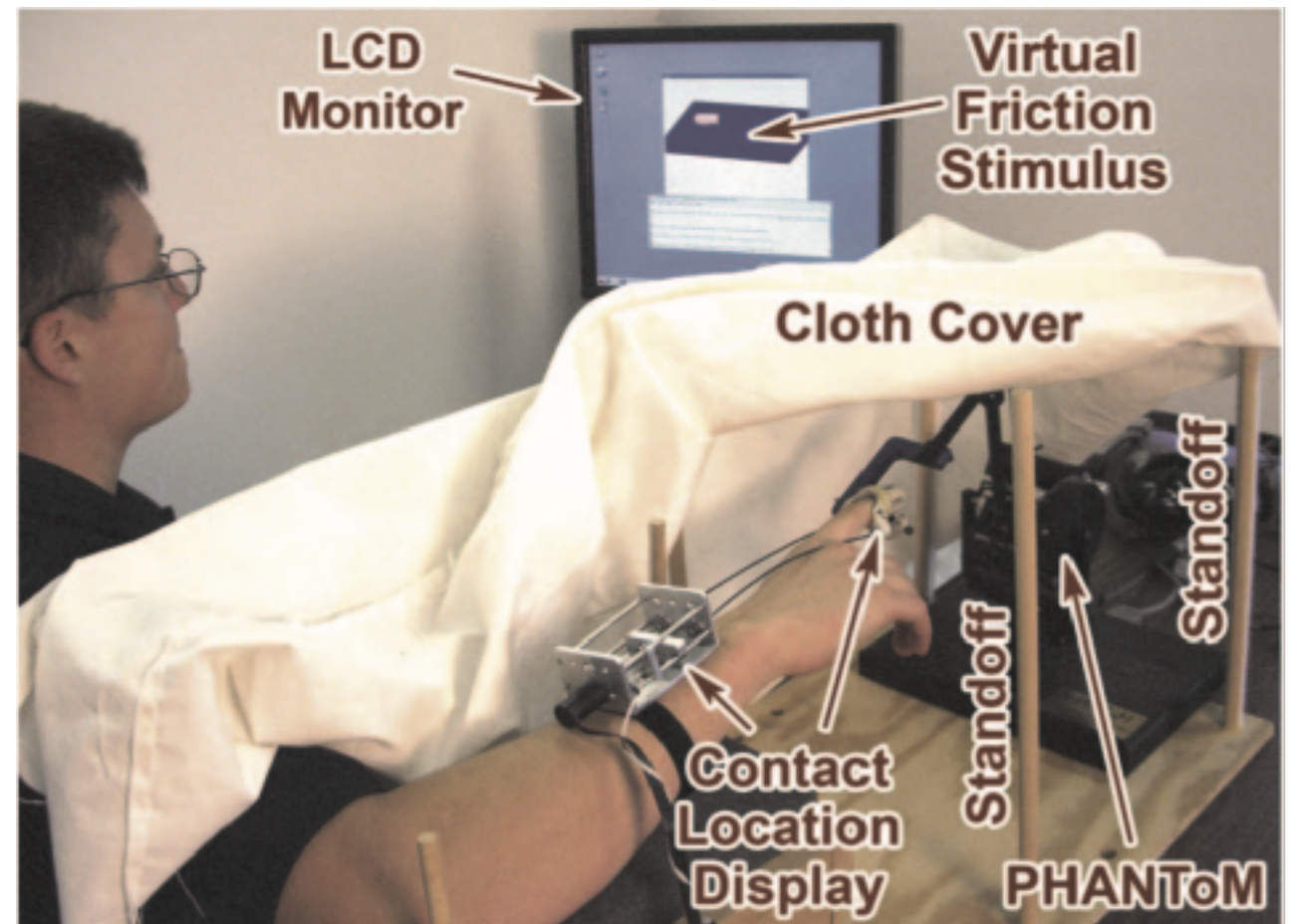
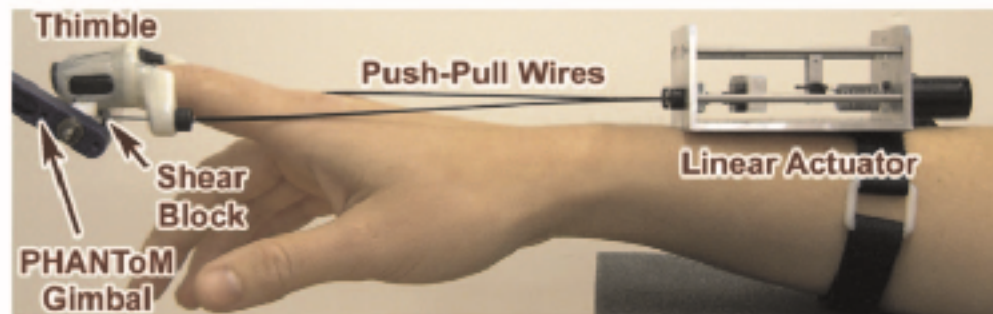
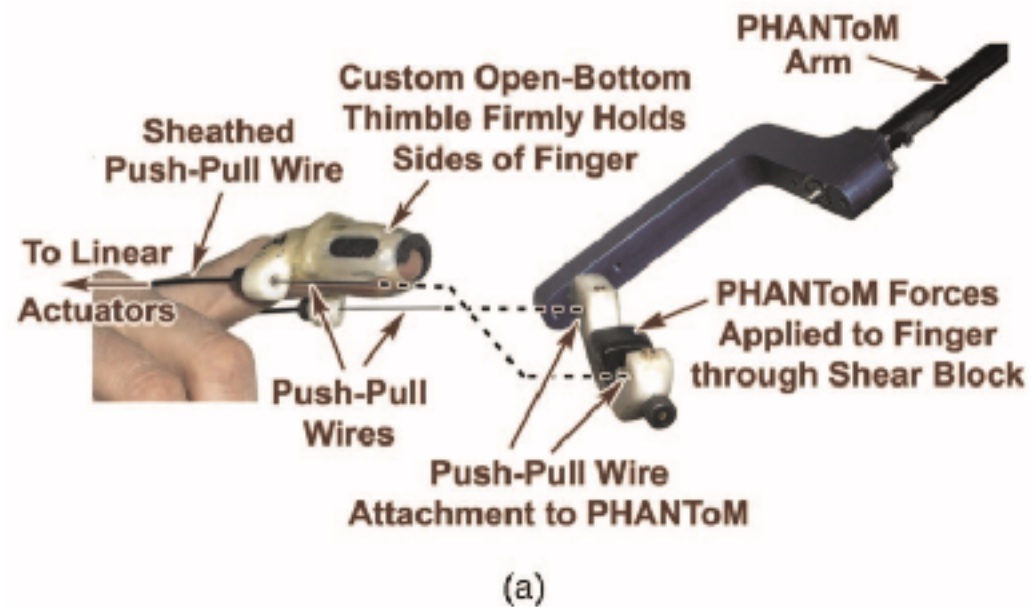
William R. Provancher, *Member, IEEE*, and Nicholas D. Sylvester

**Abstract**—This research focuses on the relative importance of fingerpad skin stretch on the perception of friction. It is hypothesized that the perceived magnitude of friction rendered by traditional force feedback can be increased through the addition of fingertip skin stretch. Perceptual data are presented from two separate tests performed on nine male subjects. The first experiment determines the perceptual thresholds for friction based on a modified Karnopp friction model where friction is rendered as purely a kinesthetic resistance via a PHANTOM force feedback device. JNDs of 0.056-0.150 corresponding to static coefficients for friction of  $\mu_s = 0.2$ -0.8 were established. The second experiment evaluates possible changes in the perceived friction magnitude due to imposing small amounts of tangential skin stretch (0.25-0.75 mm) to the fingerpad in combination with force feedback (kinesthetic resistance). Our results show that even these small amounts of skin stretch lead to a statistically significant increase in perceived friction ( $p < 0.01$ ). This significant finding will enable the hapticians to more realistically and accurately render friction via a combination of kinesthetic resistance and tactile feedback.

**Index Terms**—Tactile display, perception and psychophysics, haptic rendering, friction, skin stretch.



# Experimental setup and GFF-device:



# Constant stimuli method

Random presentation of a pre-defined set of stimuli based on:

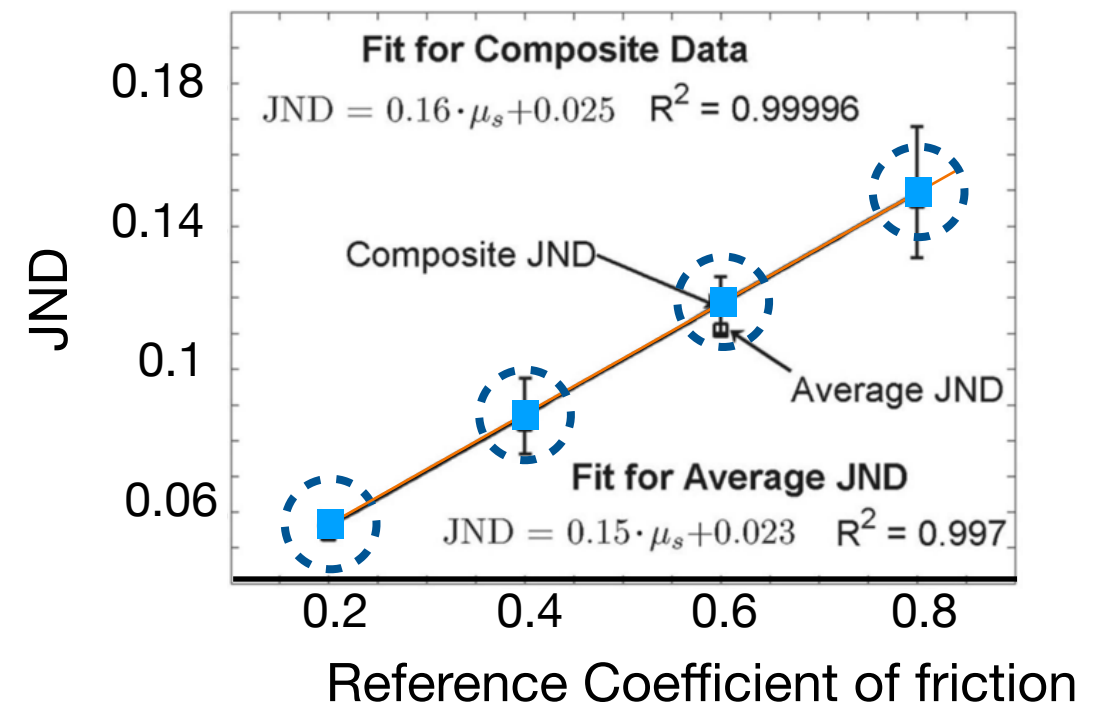
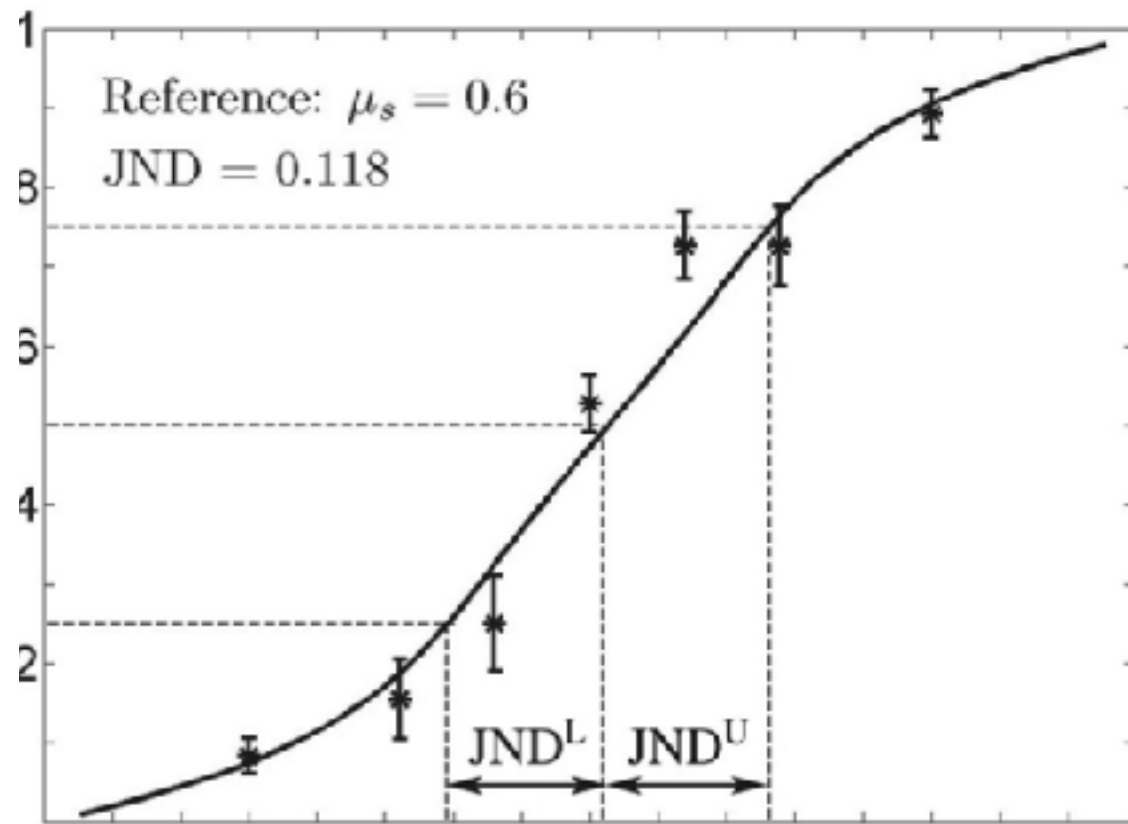
- A pre-defined set of stimuli
- Prior estimate of the sensory threshold
- Proportion of correct answers for each comparison with the reference
- Large number of repetitions per comparison (min. 10)

# Constant stimulus in our example

TABLE 1  
Experiment 1 Comparison Stimuli for Friction Levels,  
 $\mu_s = 0.2, 0.4, 0.6, 0.8$

| Reference<br>Friction | Comparison Stimuli<br>( $\mu_s$ ) |      |      |      |      |      |
|-----------------------|-----------------------------------|------|------|------|------|------|
| $\mu_s = 0.2$         | 0.02                              | 0.08 | 0.14 | 0.26 | 0.32 | 0.38 |
| $\mu_s = 0.4$         | 0.20                              | 0.27 | 0.33 | 0.47 | 0.53 | 0.60 |
| $\mu_s = 0.6$         | 0.35                              | 0.46 | 0.53 | 0.67 | 0.74 | 0.85 |
| $\mu_s = 0.8$         | 0.49                              | 0.65 | 0.72 | 0.88 | 0.95 | 1.11 |

# Results



Estimation of the probability to detect correctly the greater friction -> psychometric curve

# Scientific example of an constant method

With the same method, many key parameters for GFF devices were estimated such as the minimum force resolution

*Perception & Psychophysics*  
1991, 49 (6), 531-540

## Manual discrimination of force using active finger motion

X. D. PANG, H. Z. TAN, and N. I. DURLACH  
*Massachusetts Institute of Technology, Cambridge, Massachusetts*

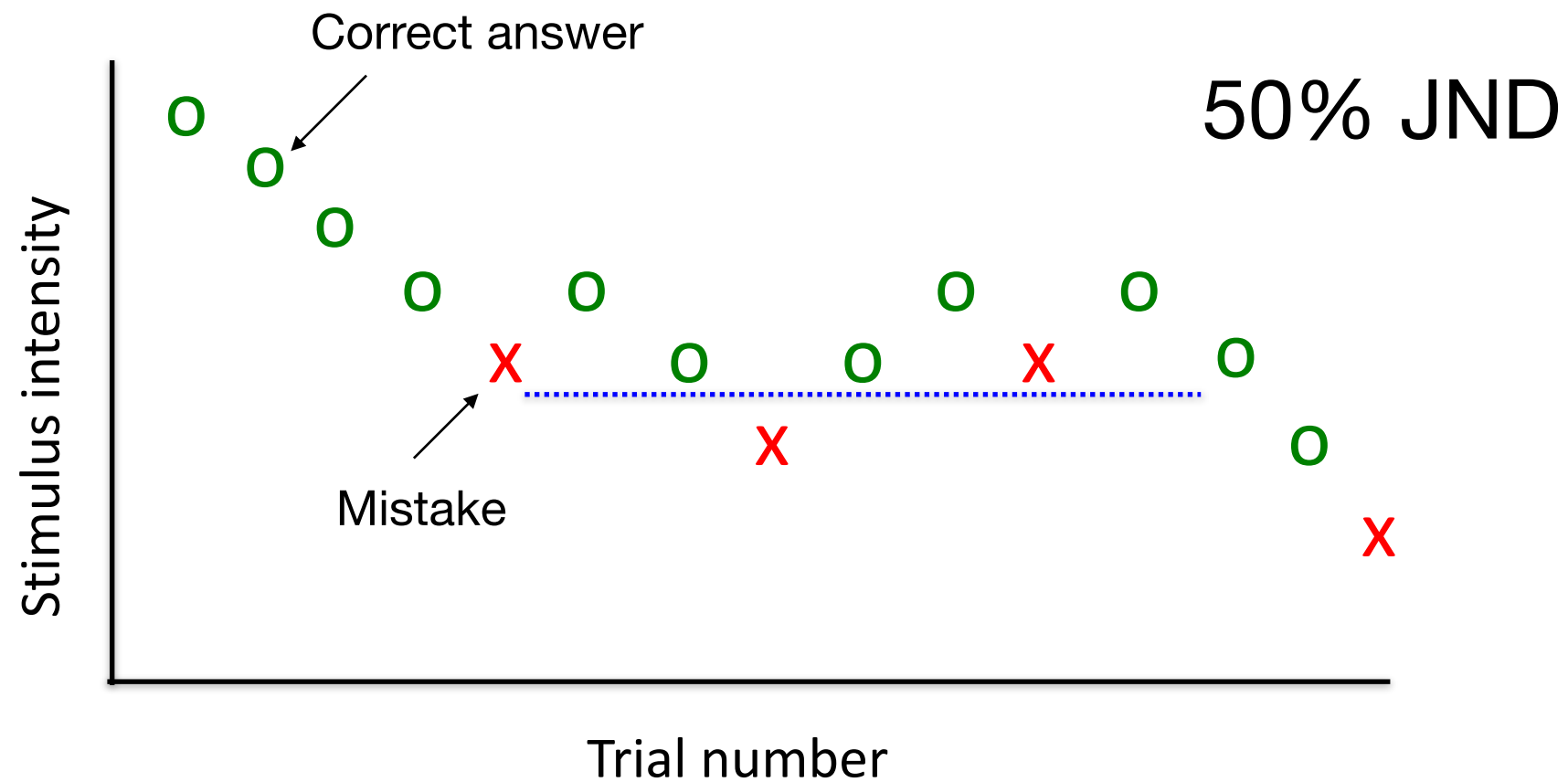
In these experiments, two plates were grasped between the thumb and forefinger and squeezed together along a linear track. An electromechanical system presented a constant resistance force during the squeeze up to a predetermined location on the track, whereupon the force effectively went to infinity (simulating a wall) or to zero (simulating a cliff). The task of the subject was to discriminate between two alternative levels of the constant resistance force (a reference level and a reference-plus-increment level). Results of these experiments indicate a just noticeable difference of roughly 7% of the reference force using a one-interval paradigm with trial-by-trial feedback over the ranges  $2.5 \leq F_0 \leq 10.0$  newtons,  $5 \leq D \leq 30$  mm,  $45 \leq S \leq 125$  mm, and  $25 \leq V \leq 160$  mm/sec, where  $F_0$  is the reference force,  $D$  is the distance squeezed,  $S$  is the initial finger-span, and  $V$  is the mean velocity of the squeeze. These results, based on tests with 5 subjects, are consistent with a wide range of previous results, some of which are associated with other body surfaces and muscle systems and many of which were obtained with different psychophysical methods.

Example for the 7% force detection

# Adaptive methods

The intensity of the stimulus is changed by successive steps depending on the previous trials in order to decrease the length of the experiment and participant's fatigue.

# Simple adaptive methods

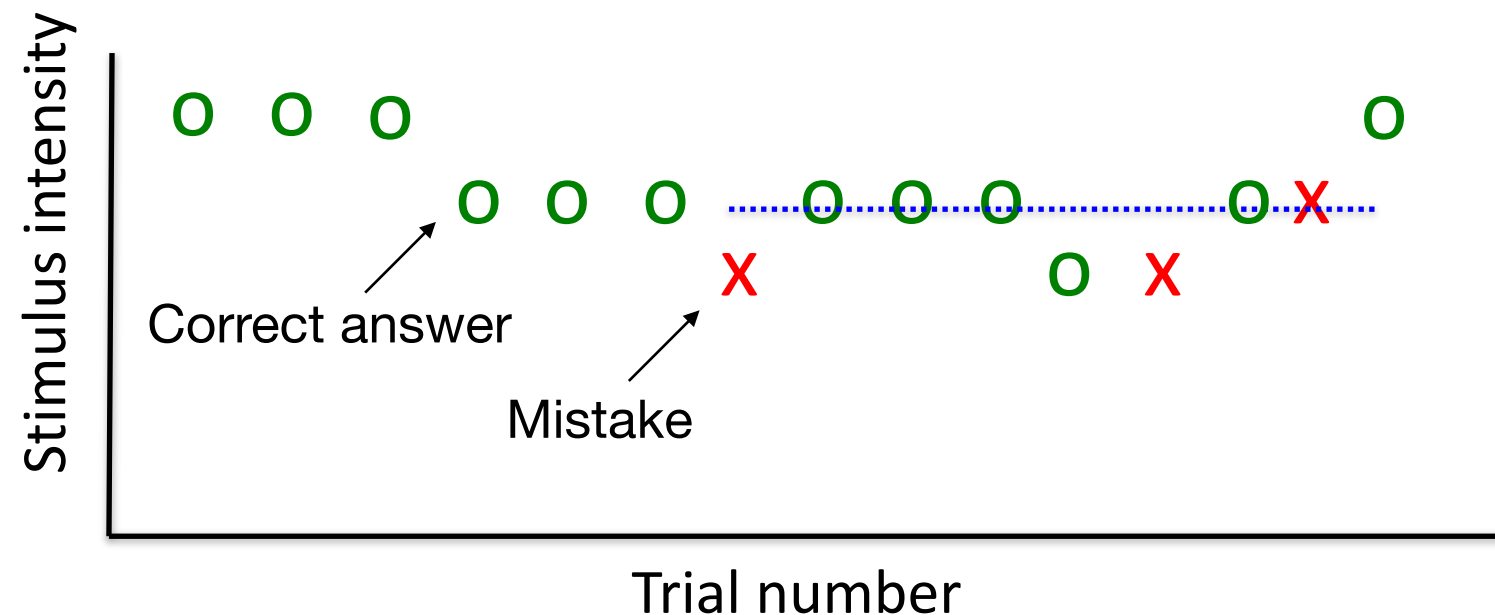


# Simple adaptive methods

Staircase methods that are converging toward a specific JND, which is not always 75%!

Many examples such as Levitt one-up three-down staircase (Levitt 1971).

The difference between the reference and the comparison stimulus is increased by a predefined step after each mistake (x) and decreased after three consecutive correct answers (o). The step is commonly reduced after the first mistake in order to fine tune the sensory threshold estimation.

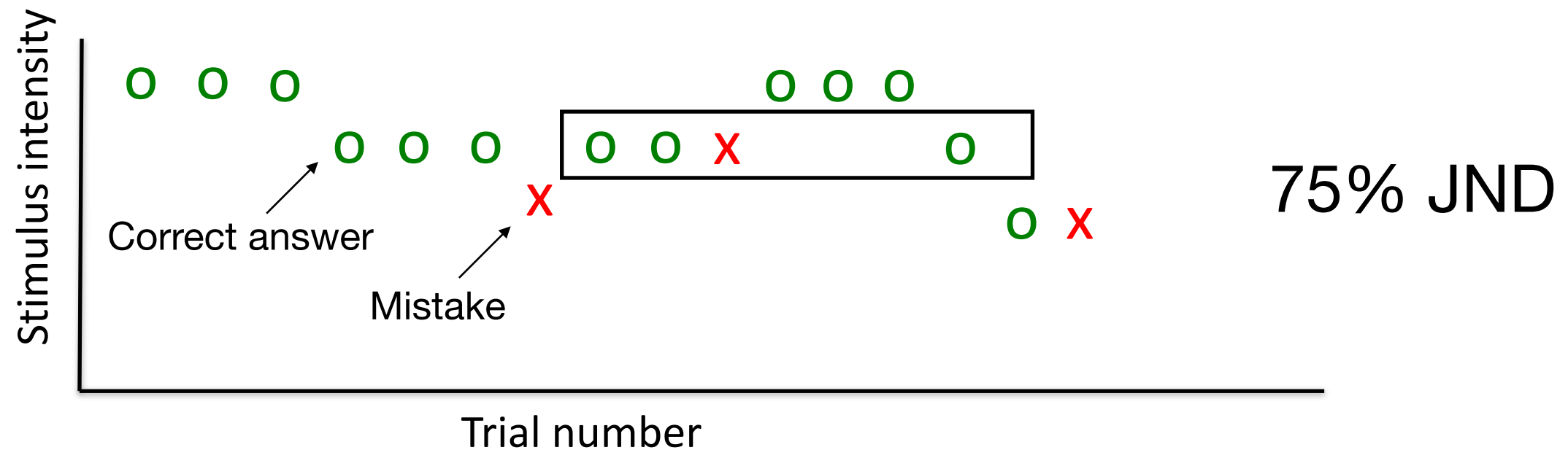


79,4% JND

The threshold is usually estimated by taking the mean of the 'peaks' and 'valleys' that occur after the first mistake.

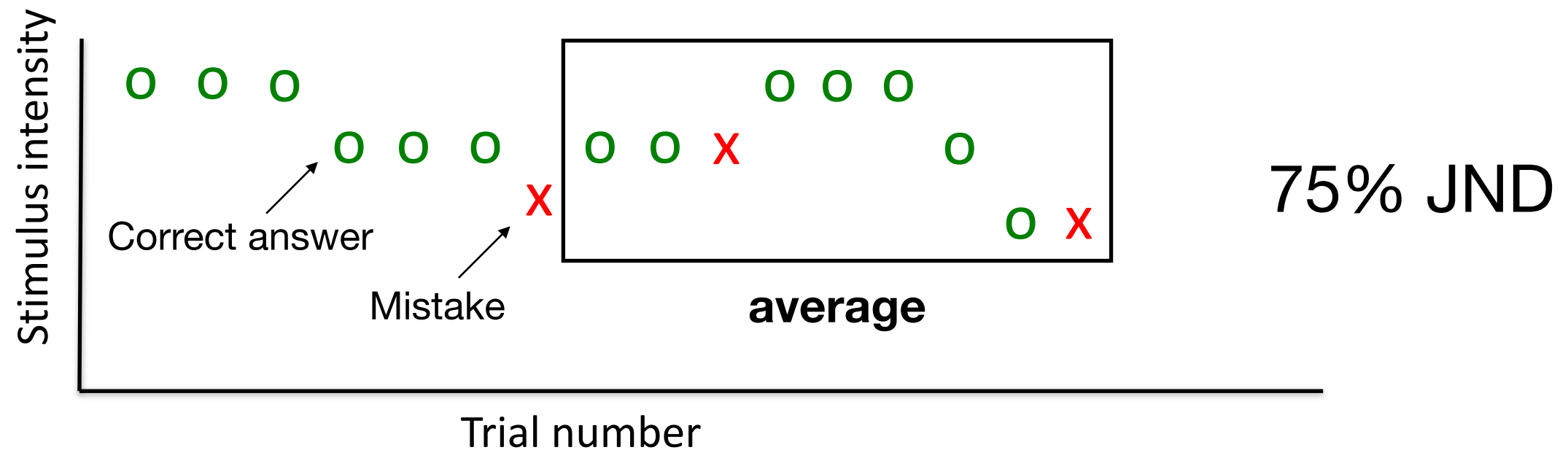
# Simple adaptive methods

The staircase can give the 75% JND if it uses non-consecutive correct answers rather than consecutive ones.



# Simple adaptive methods

The staircase can give the 75% JND if it uses non-consecutive correct answers rather than consecutive ones.

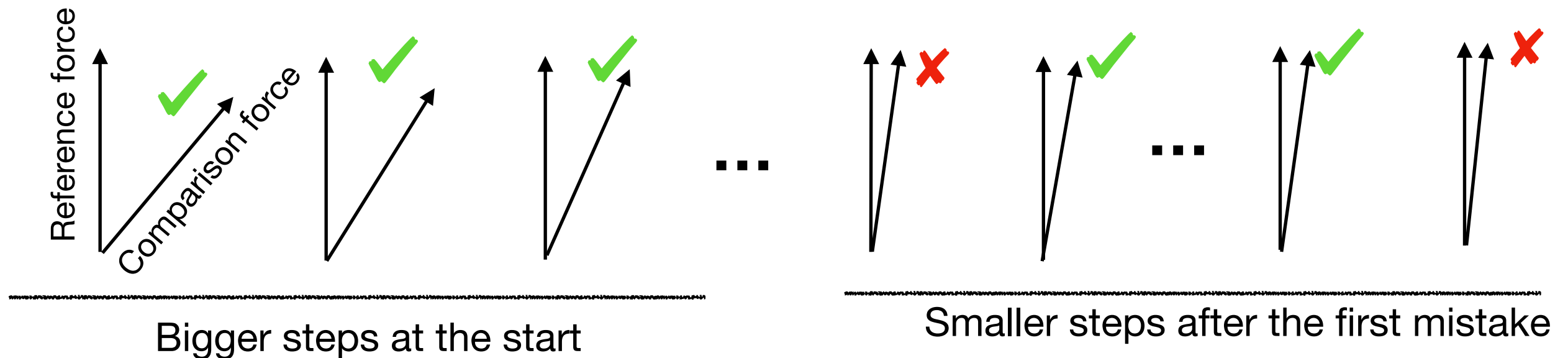


*Drawbacks of the staircase method:*

- No estimation of the full psychometric function
- Not always possible to have a constant step
- Participant can guess the staircase logic or exceed its boundaries.

# Example of adaptive method

Task: discrimination of the force direction



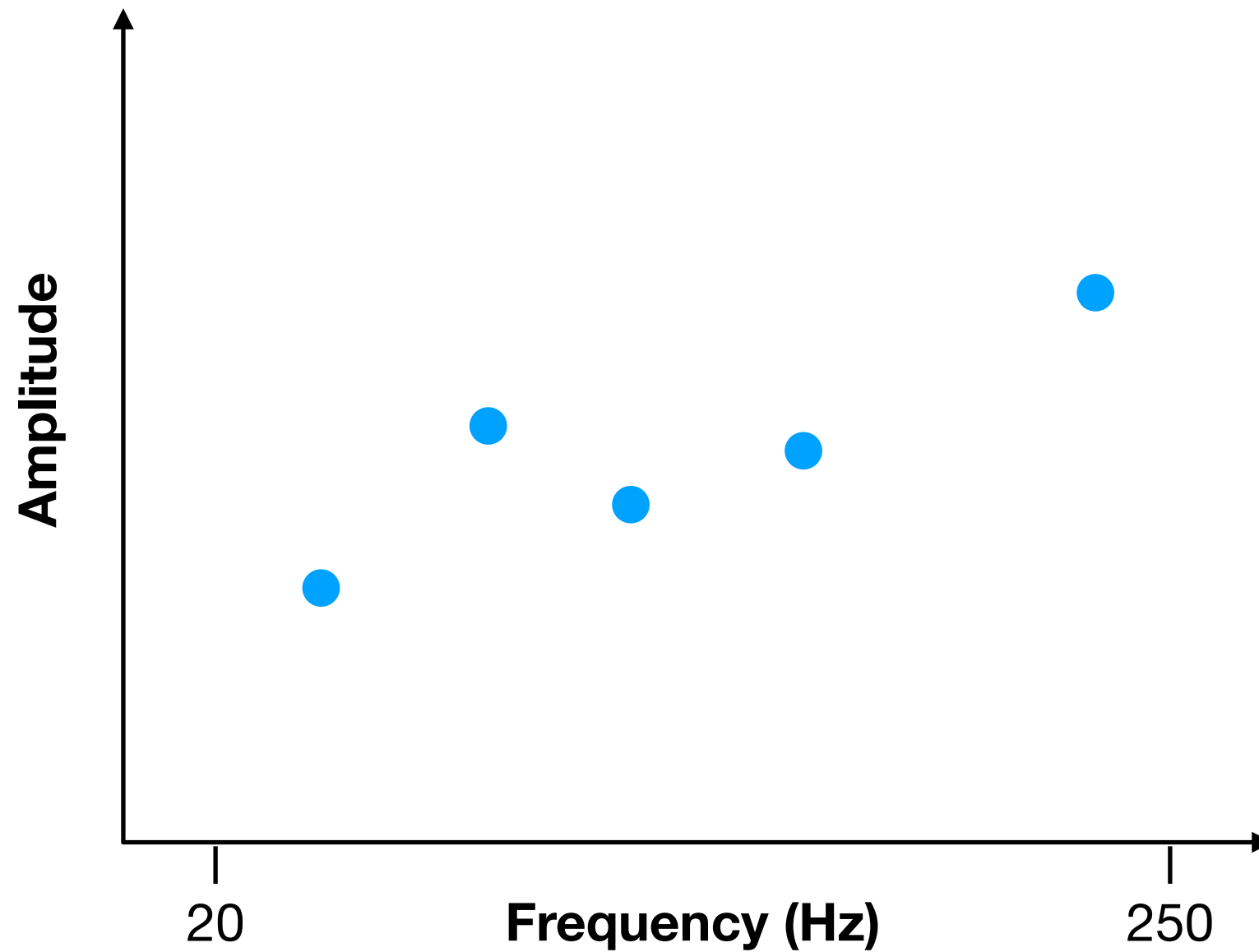
If the step size difference is not clear enough during the animation, you can make it bigger

Two references and one comparison stimulus were displayed in each trial. Participants had to report which stimulus was different from the other two.

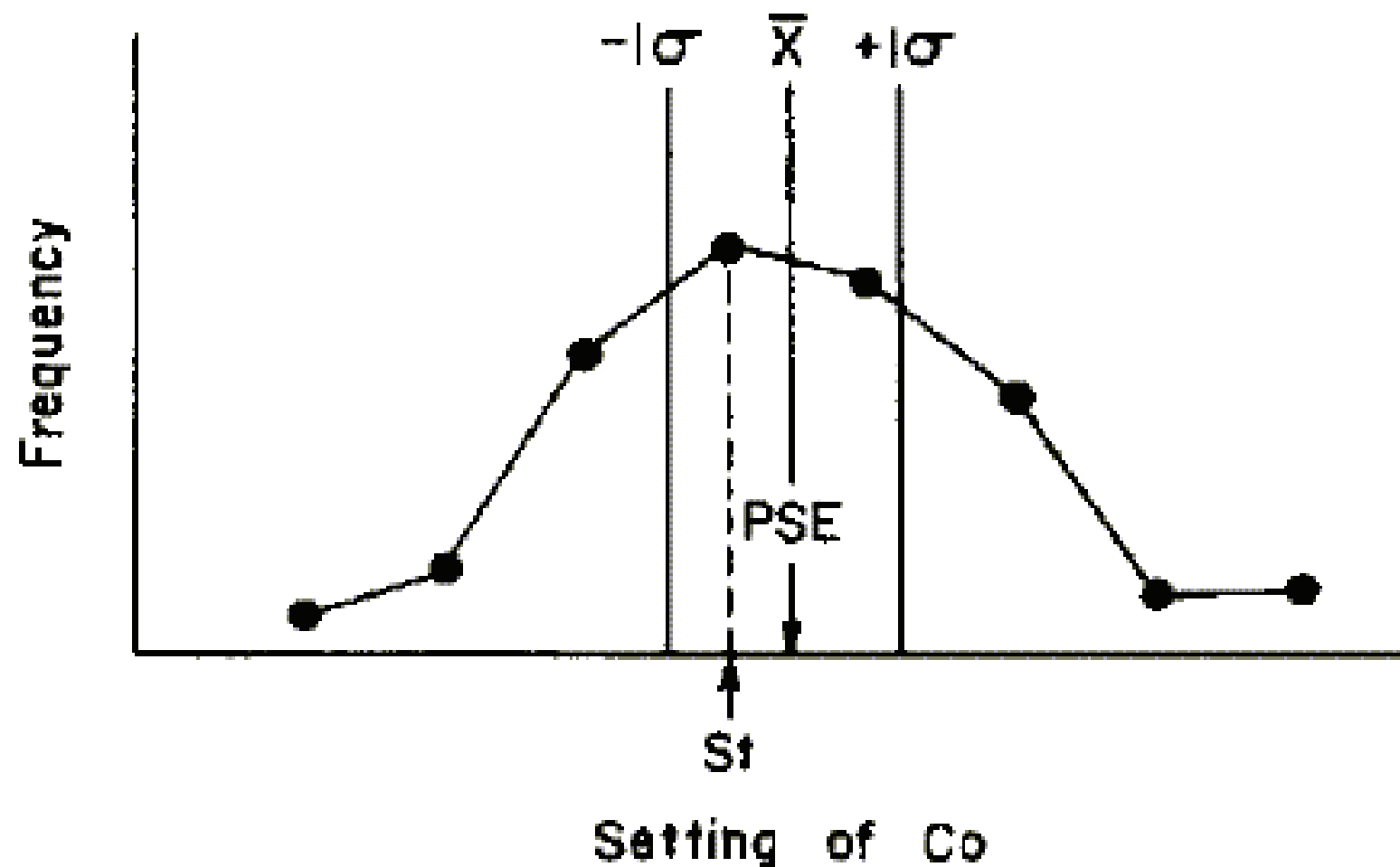
**Other psychophysical methods**

# Adjustment method

- Reference stimulus (ex: vibration at 50 Hz with amplitude A)



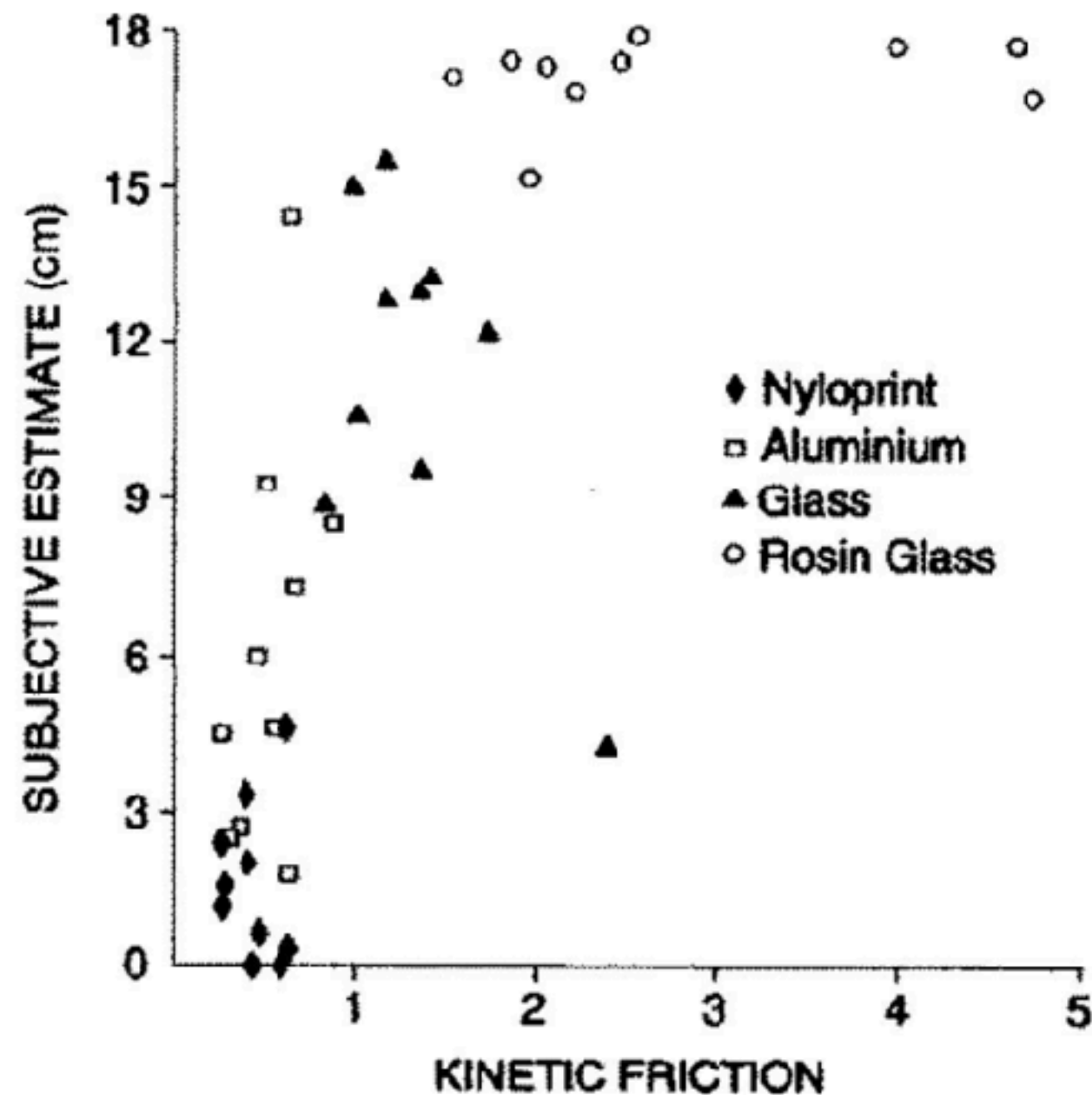
# Others: Adjustment method



**FIG. 2.8.** Frequency distribution of setting of the comparison stimulus when the method of adjustment is used to measure the difference threshold. The mean of the distribution is the point of subjective equality, and the standard deviation is used as the difference threshold.

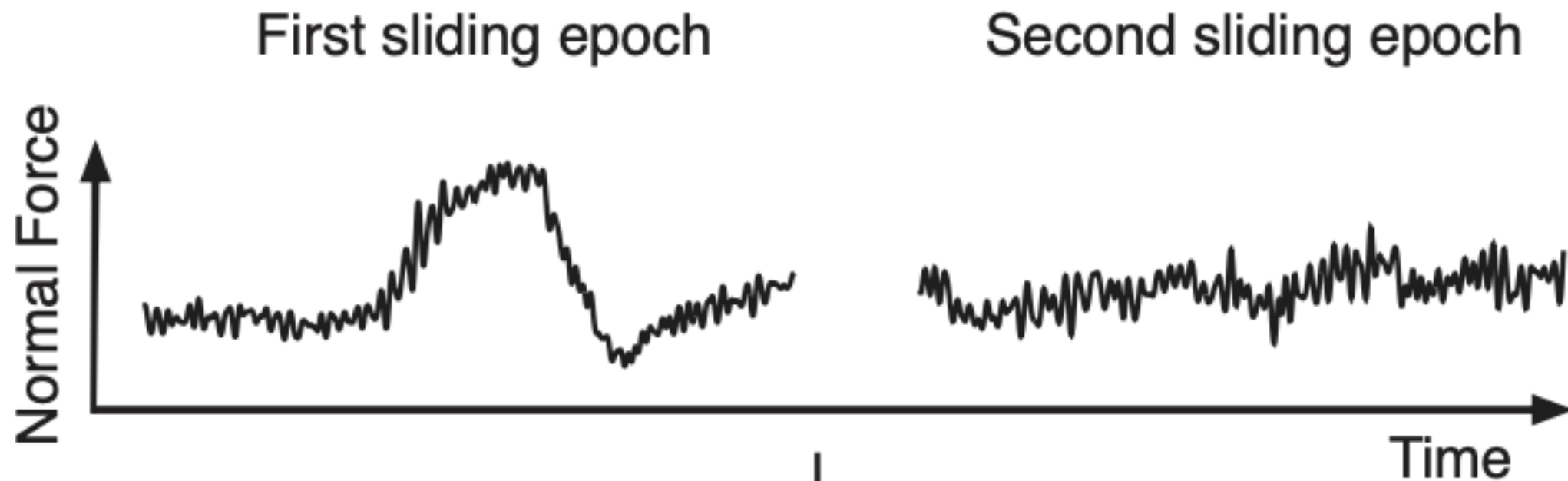
# Subjective estimates

Participant estimate the magnitude of their perception



**How to design the trials within an experiment?**

# Choice between two possibilities

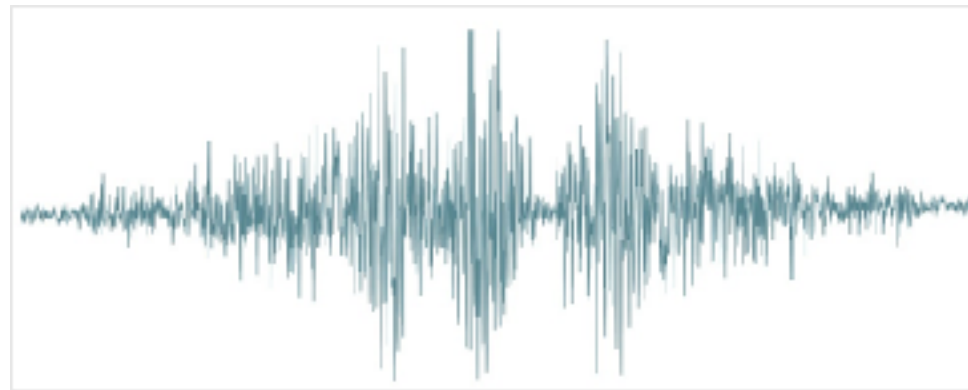


In which sliding epoch did the modulation occur?

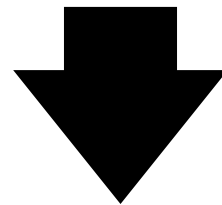
Did the normal force increase or decrease?

# Evaluation of one signal

**Trial**



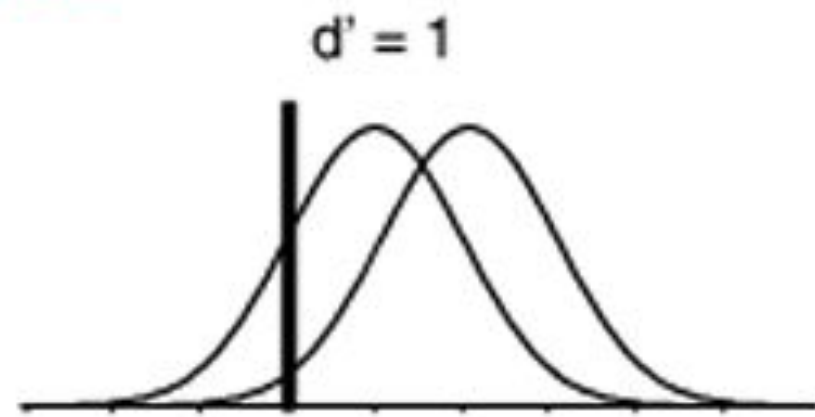
**?**



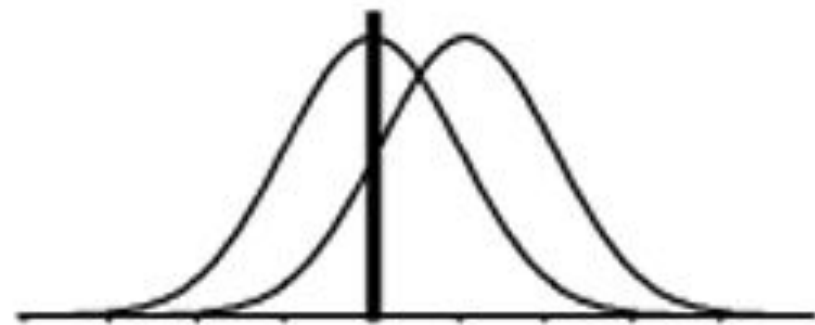
**Did I feel it ?**

**How strong it was ?**

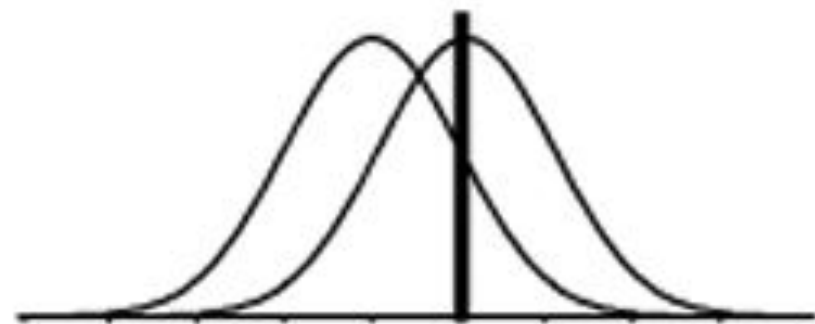
# Criterion shift



Hits = 97.5%  
False alarms = 84%



Hits = 84%  
False alarms = 50%



Hits = 50%  
False alarms = 16%

# **TP on psychophysical methods**

**08/12 & 16/12**

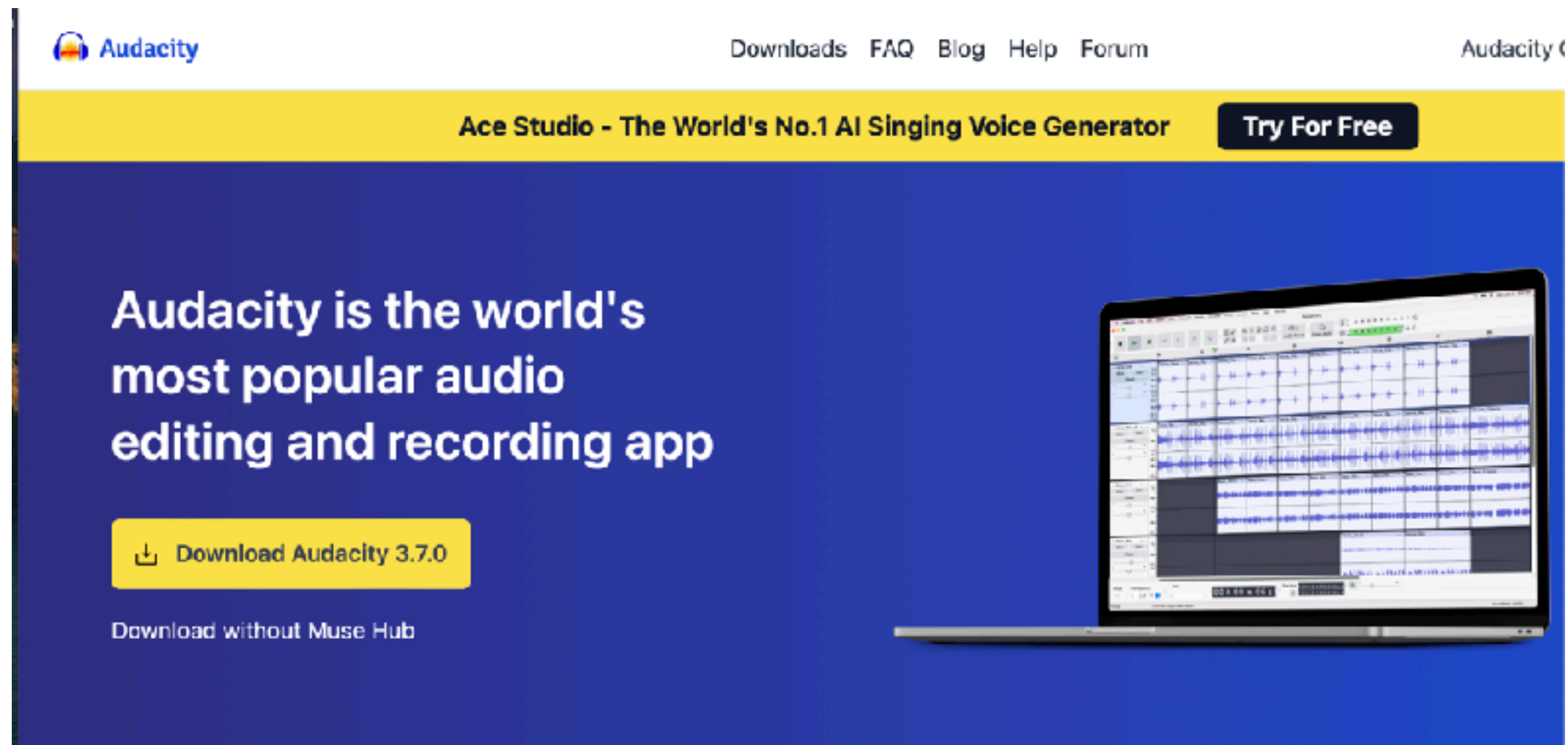
# Resources

- [https://courses.washington.edu/matlab1/Lesson\\_4.html](https://courses.washington.edu/matlab1/Lesson_4.html)
- Choi, S., & Kuchenbecker, K. J. (2012). Vibrotactile display: Perception, technology, and applications. *Proceedings of the IEEE*, 101(9), 2093-2104. (<https://ieeexplore.ieee.org/abstract/document/6353870>)
- Jones, L. A., & Tan, H. Z. (2012). Application of psychophysical techniques to haptic research. *IEEE transactions on haptics*, 6(3), 268-284.

# Organisation

08/12

- Define the groups
- Learn how to use the vibrating devices
- Find a way to create signals (Audacity, Matlab...)
- Find a study to perform



The screenshot shows the Audacity website homepage. At the top, there is a navigation bar with the Audacity logo on the left and links for Downloads, FAQ, Blog, Help, and Forum on the right. Below this is a yellow banner for 'Ace Studio - The World's No.1 AI Singing Voice Generator' with a 'Try For Free' button. The main content area has a dark blue background. On the left, it says 'Audacity is the world's most popular audio editing and recording app' in white text, followed by a yellow button that says 'Download Audacity 3.7.0'. Below this button, it says 'Download without Muse Hub'. On the right, there is an image of a laptop displaying the Audacity software interface, which shows multiple audio tracks and waveforms.

# Organisation

- Find literature on the scientific question
- Choose an experimental design
- Experiments on 4 people with two psychophysical methods
- Analyze the data in terms of JND, perceptual threshold, etc
- Prepare a presentation

# Evaluation

**Presentation on 17/12**

## **Structure of the presentation**

- Outline of the scientific question + bibliography
- Choice of the psychophysical methods
- Description of the experiments
- Results
- Discussion of the results